

Algebraic structures

Algebraic structures are sets endowed with binary operations that follow specific rules. Key examples include groups, rings, and fields, which help us understand various mathematical concepts. Homomorphisms, which are structure-preserving maps between two structured sets, play a crucial role in transferring properties across different structures. In this chapter, we will explore the importance of having a structure on a set, highlighting why structured sets are preferred over unstructured ones.

1 Binary operations

Definition 1 *Let S be a nonempty set. A binary operation on S (or an internal composition law on S) is any map*

$$\begin{aligned} f : S \times S &\longrightarrow S \\ (a, b) &\longmapsto f(a, b). \end{aligned}$$

*The image of $(a, b) \in S \times S$ is often denoted by $a * b$, $a \triangle b$, $a \oplus b$, etc., instead of $f(a, b)$. We also say that S is equipped (or endowed) with an internal composition law.*

Example 2

1. Usual addition and multiplication are binary operations on $\mathbb{N}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{Z}$, and $\mathbb{C}$.
2. Let S be a given set.
 - (a) Intersection, union, and symmetric difference are binary operations on $\mathcal{P}(S)$, the power set of S .
 - (b) The composition of maps from S into S is a binary operation on S^S , the set of all functions from S to S .
3. Let $S = \mathbb{R} \setminus \{1\}$. The operation $*$ defined by:

$$\forall x, y \in S, \quad x * y = xy - x - y + 2$$

is a binary operation on S .

Definition 3 *Let $*$ be a binary operation on a nonempty set S .*

1. The operation $*$ is said to be *commutative* if:

$$\forall x, y \in S, \quad x * y = y * x.$$

2. The operation $*$ is said to be *associative* if:

$$\forall x, y, z \in S, \quad (x * y) * z = x * (y * z).$$

3. An element $e \in S$ is called a *neutral element* if:

$$\forall x \in S, \quad x * e = e * x = x.$$

4. When S has a neutral element e for the binary operation $*$, an element $x \in S$ is said to have a *symmetric* (or *inverse*) if:

$$\exists x' \in S \text{ such that } x * x' = x' * x = e.$$

In this case, x' is called the symmetric or inverse of x .

- When S is equipped with an associative binary operation, we can perform operations in any order while maintaining the relative positions of the elements. In other words, the composition of three or more elements can be unambiguously defined, eliminating the need for parentheses.

Example 4

1. The usual operations $+$ and $\times$ defined on $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ are both associative and commutative.

2. The binary operation $*$ defined by:

$$\forall a, b \in \mathbb{R}, \quad a * b = (a + b)^2$$

is commutative but not associative.

3. The binary operation $*$ defined by:

$$\forall x, y \in \mathbb{R}, \quad x * y = |x + y| + 1$$

does not admit a neutral element.

4. The neutral element of the usual addition in $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$ is 0, and for the usual multiplication it is 1.

5. Let S be a nonempty set. The symmetric difference is a binary operation on $\mathcal{P}(S)$, which is commutative and associative.

6. The neutral element in $\mathcal{P}(S)$ for union is $\emptyset$ and for intersection is S .

7. The neutral element in $S = \mathbb{R} \setminus \{1\}$ for the binary operation $*$ defined by

$$\forall x, y \in S, \quad x * y = xy - x - y + 2$$

is 2.

8. In $\mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$, the symmetric element of an element x for the usual addition is $-x$.

9. In $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$, the symmetric of an element x different from zero for the usual multiplication is $\frac{1}{x}$.

Theorem 5 *Let $*$ be a binary operation on a nonempty set S . If a neutral element exists, it is unique.*

Suppose e_1 and e_2 are two neutral elements. Since e_1 is a neutral element we have

$$e_1 * e_2 = e_2.$$

and since e_2 is a neutral element, we also have

$$e_1 * e_2 = e_1.$$

Thus,

$$e_1 = e_2.$$

Theorem 6 *Let S be a set with an associative binary operation $*$ and admitting a neutral element e . The symmetric element of an element $x \in S$, if it exists, is unique.*

Proof. Suppose x' and x'' are two symmetric elements of x . Consider the element $x' * (x * x'')$. Given that the binary operation is associative, we have:

$$x' * (x * x'') = x' * e = x'.$$

On the other hand, we have:

$$x' * (x * x'') = (x' * x) * x'' = e * x'' = x''.$$

Combining these results, we get:

$$x' = x''.$$

■

2 Groups

The study of groups began in the post-Renaissance era, notably with Lagrange's work on solving polynomial equations around 1770. The term "group" was later introduced by Galois in 1830. Since then, group theory has become a fundamental mathematical discipline with wide-ranging applications. Its significance extends beyond mathematics to fields such as chemistry and theoretical physics. In mathematics, group theory provides a foundational framework for exploring complex structures. Its applications across various scientific domains underscore its universality and profound impact on our understanding of symmetry, transformations, and relationships within complex systems.

Definition 7 Let G be a nonempty set equipped with a binary operation $*$. Then G is said to be a group if:

1. The operation $*$ is associative,
 2. G contains a neutral element e with respect to $*$,
 3. Every element of G possesses an inverse element with respect to $*$.
- A group G with a binary operation $*$ is denoted by $(G, *)$. When there is no ambiguity, it is simply denoted as G .
 - If the binary operation $*$ is also commutative, we say that $(G, *)$ is an abelian or *commutative group*.

Example 8

1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$, and $(\mathbb{C}, +)$ are abelian groups.
2. $(\mathbb{Q}^*, \times)$, $(\mathbb{R}^*, \times)$, and $(\mathbb{C}^*, \times)$ are abelian groups.
3. $(\{+1, -1\}, \times)$ is an abelian group.
4. Let D be the set of decimal numbers. $(D, +)$ is an abelian group. Recall that a decimal number is any rational that can be written as a fraction whose denominator is a power of 10.
5. Let E be a given set and the set of bijective maps from E into E , and let $\circ$ denote the composition of maps. Then $(G, \circ)$ is a noncommutative group.

Notation

- Sometimes we denote a binary operation on a group G by $(+)$. This notation is called additive. In this case, the symmetric element of an element $x \in G$ is denoted $-x$, and for any $n \in \mathbb{Z}$, we define:

$$nx = \begin{cases} e & \text{if } n = 0, \\ \underbrace{x + x + \cdots + x}_{n \text{ times}} & \text{if } n > 0, \\ \underbrace{(-x) + (-x) + \cdots + (-x)}_{-n \text{ times}} & \text{if } n < 0. \end{cases}$$

- Sometimes we denote a binary operation on a group G by $(\times)$ or $(\cdot)$. The notation $(G, \cdot)$ is called multiplicative. In this notation, the symmetric element of an element $x \in G$ is denoted x^{-1} , and for any $n \in \mathbb{Z}$ we define:

$$x^n = \begin{cases} e & \text{if } n = 0, \\ \underbrace{x \cdot x \cdots x}_{n \text{ times}} & \text{if } n > 0, \\ \underbrace{x^{-1} x^{-1} \cdots x^{-1}}_{-n \text{ times}} & \text{if } n < 0. \end{cases}$$

- In mathematics, it is common practice to omit multiplication sign when the context is clear and does not lead to ambiguity. So we write ab instead of $a \times b$ or $a \cdot b$.

When we adopt multiplicative notation, we easily prove that, for all $x, y \in G$, and $n, m \in \mathbb{Z}$, we have the following properties:

1. $(x^{-1})^{-1} = x$.
2. $(xy)^{-1} = y^{-1}x^{-1}$.
3. $x^n x^m = x^{n+m}$.
4. $(x^n)^m = x^{nm}$.
5. $(x^n)^{-1} = (x^{-1})^n$.

- Note that if a group G is commutative, it is customary to express the group operation using additive notation. In this case, analogous results to those mentioned above apply.

2.1 Direct product of groups

- Let G_1 and G_2 be two groups. The binary operation defined on

$$G_1 \times G_2 = \{(x, y) : x \in G_1, y \in G_2\}$$

by

$$(x_1, y_1)(x_2, y_2) = (x_1 x_2, y_1 y_2)$$

equips this set with a group structure. Note that the neutral element is (e_{G_1}, e_{G_2}) , and the inverse of an element $(x, y) \in G_1 \times G_2$ is (x^{-1}, y^{-1}) .

- In a similar way, when $G_1, G_2, \dots, G_n$ are groups, we define a group structure on $G_1 \times G_2 \times \cdots \times G_n$ via the binary operation defined by:

$$\forall (x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in G_1 \times G_2 \times \cdots \times G_n : (x_1, x_2, \dots, x_n)(y_1, y_2, \dots, y_n) = (x_1 y_1, x_2 y_2, \dots, x_n y_n).$$

Example 9 For any positive integer n , the sets: $\mathbb{Z}^n, \mathbb{Q}^n, \mathbb{R}^n$, and $\mathbb{C}^n$ can be endowed with canonical group structures obtained from the structure of group of $(\mathbb{Z}, +), (\mathbb{Q}, +), (\mathbb{R}, +)$, and $(\mathbb{C}, +)$, respectively.

2.2 Subgroups

Definition 10 Let G be a group and H a subset of G . We say that H is a subgroup of G if:

1. H is nonempty,
2. $\forall x, y \in H, xy \in H$ (i.e., H is stable under the binary operation),
3. $\forall x \in H, x^{-1} \in H$.

It is easy to show that conditions 2) and 3) can be replaced by the condition: $\forall x, y \in H, xy^{-1} \in H$. The previous definition is equivalent to saying that the restriction of the binary operation on G to H defines a binary operation on H , making H itself a group.

Often, to show that a set equipped with a binary operation is a group, it is easier, to show that it is a subgroup of a known group.

Example 11

1. For any group G , the sets G and $\{e\}$ are subgroups of G (called the trivial subgroups).
2. $\mathbb{Z}$ is a subgroup of $(\mathbb{Q}, +)$, which is a subgroup of $(\mathbb{R}, +)$, which is a subgroup of $(\mathbb{C}, +)$.
3. Let $\alpha \in \mathbb{R}$ and $H = \{\alpha k : k \in \mathbb{Z}\}$. Then H is a subgroup of $(\mathbb{R}, +)$.
4. Let $a \in \mathbb{R}^*$. The set $U_a = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of the group $(\mathbb{R}^*, \times)$.
5. $\mathbb{R}_+^*$ is a subgroup of $(\mathbb{R}^*, \times)$.
6. The set $U = \{z \in \mathbb{C} : |z| = 1\}$ is a subgroup of $(\mathbb{C}^*, \times)$.
7. Let $n \in \mathbb{N}^*$. The set $U_n = \{z \in \mathbb{C} : z^n = 1\}$ is a subgroup of the group U .

Theorem 12 Let H and K be two subgroups of a group G . Then $H \cap K$ is a subgroup of G .

Proof. Suppose H and K are subgroups of G . Then $e_G \in H$ and $e_G \in K$, so $e_G \in H \cap K$. Now, let $x, y \in H \cap K$. Then $x, y \in H$ and $x, y \in K$. Since H and K are subgroups, it follows that $xy^{-1} \in H$ and $xy^{-1} \in K$, hence $xy^{-1} \in H \cap K$. Therefore, $H \cap K$ is a subgroup of G . ■

► Beware that the union of two subgroups is not generally a subgroup. Indeed,

$H_1 = 2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$ and $H_2 = 3\mathbb{Z} = \{3k : k \in \mathbb{Z}\}$ are subgroups of $(\mathbb{Z}, +)$. But their union $H_1 \cup H_2$ is not a subgroup of $\mathbb{Z}$. This is because $2, 3 \in H_1 \cup H_2$ and $2 + 3 = 5 \notin H_1 \cup H_2$.

Proposition 13 *Let H and K be two subgroups of a group G . Then $H \cup K$ is a subgroup of G if and only if $H \subset K$ or $K \subset H$.*

Proof. Let H and K be subgroups of a group G . It is clear that if one subgroup is contained in the other, then their union is a subgroup of G . Conversely, suppose $H \cup K$ is a subgroup of G and that $H \not\subset K$ and $K \not\subset H$, i.e., there exists $x \in H$ with $x \notin K$ and there exists $y \in K$ with $y \notin H$. Now, $xy \in H \cup K$, so $xy \in H$ or $xy \in K$. If $xy \in H$, then $x^{-1}xy = y \in H$, which is a contradiction because $y \notin H$. Similarly, if $xy \in K$, then $xyy^{-1} = x \in K$, which is a contradiction because $x \notin K$. ■

Proposition 14 *A subset H of $\mathbb{Z}$ is a subgroup of $(\mathbb{Z}, +)$ if and only if $H = n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for some $n \in \mathbb{N}_0$.*

Proof. Firstly, it is clear that for any $n \in \mathbb{N}$, the subset

$$n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$$

is a subgroup of $\mathbb{Z}$. Conversely, let H be a subgroup of $(\mathbb{Z}, +)$. If $H = \{0\}$, then $H = 0\mathbb{Z}$. Now assume $H \neq \{0\}$. Then there exists $x \in H$ with $x \neq 0$. Since H is a subgroup, it contains at least one nonzero element of $\mathbb{N}_0$. Let

$$\Gamma = H \cap \mathbb{N}.$$

We have $\Gamma \neq \{0\}$. Since Γ is a subset of $\mathbb{N}$, it contains a smallest nonzero element, which we denote by n . We will show that $H = n\mathbb{Z}$. The stability under addition ensures that

$$n\mathbb{Z} \subseteq H.$$

Let $x \in H$. By the Euclidean algorithm, there exist $q \in \mathbb{Z}$ and $r \in \{0, 1, \dots, n-1\}$ such that

$$x = qn + r.$$

Since $x \in H$ and H is stable under addition,

$$r = x - qn \in H.$$

Because n is minimal, we must have $r = 0$.

Thus, $x \in n\mathbb{Z}$. This concludes that

$$H = n\mathbb{Z}.$$

■

- It is clear that any subgroup of an abelian group is itself abelian. We will show later in an example that a non-abelian group can contain abelian subgroups as well as non-abelian subgroups.

2.3 Group homomorphisms

In abstract algebra, a homomorphism is a structure-preserving map between two algebraic structures, such as groups:

Definition 15 Let $(G, *)$ and (G', Δ) be two groups. A map $f : G \rightarrow G'$ is called a group homomorphism (or simply group morphism) if

$$\forall a, b \in G, f(a * b) = f(a) \Delta f(b).$$

- Strictly speaking, we should always distinguish between the binary operations of G and G' . However, the context generally makes the distinction clear without having to explicitly adopt different notations.

Definition 16 Let $f : G \rightarrow G'$ be a group homomorphism.

1. f is called an isomorphism if f is bijective.
2. f is called a monomorphism if f is injective.
3. f is called an endomorphism if $G = G'$ (i.e., G and G' are the same set equipped with the same binary operation).
4. f is called an epimorphism if f is surjective.
5. f is called an automorphism if f is a bijective endomorphism.

We denote $\text{Hom}(G, G')$ as the set of homomorphisms from G into G' , $\text{Isom}(G, G')$ as the set of bijective homomorphisms from G into G' , $\text{End}(G)$ as the set of endomorphisms of G , $\text{Aut}(G)$ as the set of automorphisms of G , $\text{Epi}(G, G')$ as the set of epimorphisms from G to G' , $\text{Mon}(G, G')$ as the set of monomorphisms from G into G' .

Example 17

1. For any groups G and G' , the constant map defined by

$$\begin{array}{ccc} f : G & \longrightarrow & G' \\ g & \longmapsto & e_{G'} \end{array}$$

is a group homomorphism.

2. For every group G , the identity map

$$\begin{array}{ccc} id_G : G & \longrightarrow & G \\ g & \longmapsto & g \end{array}$$

is a group automorphism.

3. Let G be a group and $a \in G$. Then the map defined as

$$\begin{aligned} f : \mathbb{Z} &\longrightarrow G \\ n &\longmapsto a^n \end{aligned}$$

is a homomorphism from $(\mathbb{Z}, +)$ into $(G, \cdot)$.

4. The complex exponential map

$$\begin{aligned} f : \mathbb{C} &\longrightarrow \mathbb{C}^* \\ z &\longmapsto e^{2\pi iz} \end{aligned}$$

is a group homomorphism from $(\mathbb{C}, +)$ into $(\mathbb{C}^*, \times)$.

5. The map defined as

$$\begin{aligned} f : \mathbb{C}^* &\longrightarrow \mathbb{R}^* \\ z &\longmapsto |z| \end{aligned}$$

is a monomorphism from $(\mathbb{C}^*, \times)$ into $(\mathbb{R}^*, \times)$.

6. For any element g in a group G , the map defined as

$$\begin{aligned} h : G &\longrightarrow G \\ a &\longmapsto gag^{-1} \end{aligned}$$

is a group automorphism (called the conjugation homomorphism).

7. Let G_1 and G_2 be two groups. The binary operation defined on $G_1 \times G_2 = \{(x, y) : x \in G_1, y \in G_2\}$ by

$$(x_1, y_1)(x_2, y_2) = (x_1x_2, y_1y_2)$$

equips this set with a group structure. Moreover, the two canonical projections

$$\begin{aligned} \rho_1 : G_1 \times G_2 &\longrightarrow G_1 \\ (x_1, y_1) &\longmapsto x_1 \end{aligned}$$

and

$$\begin{aligned} \rho_2 : G_1 \times G_2 &\longrightarrow G_2 \\ (x_1, y_1) &\longmapsto y_1 \end{aligned}$$

are group homomorphisms from $G_1 \times G_2$ into G_1 and G_2 , respectively.

Proposition 18 Let $f : G \rightarrow G'$ be a group homomorphism. The following properties hold:

1. $f(e_G) = e_{G'}$, where e_G and $e_{G'}$ denote the neutral elements of G and G' , respectively.
2. For all $x \in G$ and all $n \in \mathbb{Z}$, we have $f(x^n) = (f(x))^n$.

3. If K is a subgroup of G' , then $f^{-1}(K)$ is a subgroup of G .
4. If H is a subgroup of G , then $f(H)$ is a subgroup of G' .
5. The kernel of f , defined as $\ker f = \{x \in G : f(x) = e_{G'}\}$, is a subgroup of G .
6. The image of f , defined as $\text{Im } f = \{y \in G' : \exists x \in G, f(x) = y\}$, is a subgroup of G' .

Proof.

1. Since f is a homomorphism, we have

$$\begin{aligned} f(e_G) &= f(e_G e_G) \\ &= f(e_G) f(e_G). \end{aligned}$$

Composing both sides with $(f(e_G))^{-1}$ gives

$$\begin{aligned} e_{G'} &= (f(e_G))^{-1} f(e_G) \\ &= f(e_G). \end{aligned}$$

Thus, $f(e_G) = e_{G'}$.

2. When $n \geq 0$ we use a proof by induction on n . For $n = 0$. We know that by convention $x^0 = e_G$. Then

$$f(x^0) = f(e_G) = e_{G'} = (f(x))^0.$$

Assume the property holds for n , and show that it holds for $n + 1$. We have

$$\begin{aligned} f(x^{n+1}) &= f(x^n x) \\ &= f(x^n) f(x) \\ &= (f(x))^n f(x) \\ &= (f(x))^{n+1}. \end{aligned}$$

For $n < 0$, consider $-n$. Since $f(x^n) = (f(x))^n$, we have $f(x^{-n}) = (f(x))^{-n}$ and $f(x^n) = (f(x))^n$. By induction, we can extend this to all integers n .

3. Let K be a subgroup of G' . We know that $e_{G'} \in K$ and $f(e_G) = e_{G'}$. Thus, $e_G \in f^{-1}(K)$. Let $x, y \in f^{-1}(K)$. Then $f(x), f(y) \in K$. Since K is a subgroup, $f(x)(f(y))^{-1} = f(xy^{-1}) \in K$. Therefore, $xy^{-1} \in f^{-1}(K)$. Hence, $f^{-1}(K)$ is a subgroup of G .
4. Let H be a subgroup of G . Since $e_G \in H$, then $f(e_G) = e_{G'} \in f(H)$. Let $x', y' \in f(H)$. Then $x' = f(x)$ and $y' = f(y)$ for some $x, y \in H$, and we have

$$(y')^{-1} = (f(y))^{-1} = f(y^{-1}).$$

As f is a homomorphism of groups we have

$$x'(y')^{-1} = f(x)f(y^{-1}) = f(xy^{-1})$$

and

$$f(xy^{-1}) \in f(H)$$

because H is a subgroup and $x, y \in H$. We conclude that $f(H)$ is a subgroup of G' .

5. This follows directly from property (3).
6. This follows directly from property (4).

■

Proposition 19 *A group homomorphism $f : G \rightarrow G'$ is injective if and only if $\ker f = \{e_G\}$.*

Proof. Let $f : G \rightarrow G'$ be a group homomorphism. Assume that f is injective. Let $x \in \ker f$. By definition of the kernel,

$$f(x) = e_{G'}.$$

Since f is injective, $f(x) = f(e_G)$ implies $x = e_G$. Therefore,

$$\ker f = \{e_G\}.$$

Conversely, suppose $\ker f = \{e_G\}$. Let $x, y \in G$ with $f(x) = f(y)$. This gives

$$f(x)f(y)^{-1} = f(y)f(y)^{-1}.$$

Then

$$f(xy^{-1}) = e_{G'}.$$

Thus, $xy^{-1} \in \ker f$. Given that $\ker f = \{e_G\}$, we have

$$xy^{-1} = e_G,$$

which implies $x = y$. Hence, f is injective. ■

Proposition 20 *Let G , G' , and G'' be groups.*

1. *If f is a homomorphism from G into G' and g is a homomorphism from G' into G'' , then $g \circ f$ is a homomorphism from G into G'' .*
2. *If $f : G \rightarrow G'$ is an isomorphism, then its inverse $f^{-1} : G' \rightarrow G$ is also an isomorphism.*

Proof.

1. Let f be a homomorphism from G into G' and g a homomorphism from G' into G'' . For any $x, y \in G$, we have

$$(g \circ f)(xy) = g(f(xy)).$$

Since f is a homomorphism, $f(xy) = f(x)f(y)$. Therefore,

$$g(f(xy)) = g(f(x)f(y)).$$

Since g is also a homomorphism, $g(f(x)f(y)) = g(f(x))g(f(y))$. Thus,

$$(g \circ f)(xy) = g(f(x))g(f(y)) = (g \circ f)(x)(g \circ f)(y).$$

Hence, $g \circ f$ is a homomorphism from G into G'' .

2. Suppose $f : G \rightarrow G'$ is an isomorphism. Let $h = f^{-1} : G' \rightarrow G$. We know that h is bijective. We need to show that h is a homomorphism. Let $x, y \in G'$. Put $x = f(a)$ and $y = f(b)$ for some $a, b \in G$. Since f is a homomorphism, we have

$$f(a)f(b) = f(ab).$$

Thus,

$$h(xy) = h(f(a)f(b)) = h(f(ab)) = ab.$$

Since $h(x) = a$ and $h(y) = b$, we get

$$h(xy) = ab = h(x)h(y).$$

Therefore, h is a homomorphism. Since h is bijective and a homomorphism, it is an isomorphism.

■

2.4 Quotient group

Let $(G, +)$ be an abelian group and let H be a subgroup of G . Consider the binary relation $\mathcal{R}$ defined on G by:

$$\forall x, y \in G, x\mathcal{R}y \text{ if and only if } x - y \in H.$$

This relation $\mathcal{R}$ is an equivalence relation. We denote the quotient set $G/\mathcal{R}$ by G/H .

We define a binary operation on G/H as follows:

$$\forall \bar{x}, \bar{y} \in G/H : \bar{x} \oplus \bar{y} = \overline{x + y}.$$

Then $(G, \oplus)$ is a group called *quotient group*. Indeed, the associativity of $\oplus$ follows from the associativity of $+$ in G . The neutral element is $\bar{e}$, where e is the neutral element of G , and the symmetric element of an element $\bar{x}$ is $\overline{-x}$, where $-x$ is the symmetric of x in G .

Example 21 For any nonnegative integer n , $n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$. Let $\mathcal{R}$ be the equivalence relation defined on $\mathbb{Z}$ by:

$$x\mathcal{R}y \text{ if and only if } x - y \in n\mathbb{Z}.$$

The equivalence class of $x \in \mathbb{Z}$ is denoted by $\bar{x}$.

The quotient set of $\mathbb{Z}$ by $\mathcal{R}$ is:

$$\mathbb{Z}/\mathcal{R} = \{\bar{x} : x \in \mathbb{Z}\} = \{r : r \in \{0, 1, \dots, n-1\}\},$$

which is denoted by $\mathbb{Z}/n\mathbb{Z}$. It is clear that $\mathbb{Z}/n\mathbb{Z}$ is finite, and its cardinality is n . It is immediate to verify that $\mathcal{R}$ is compatible with both addition and multiplication on $\mathbb{Z}$. Therefore, the quotient $\mathbb{Z}/n\mathbb{Z}$ is canonically equipped with two laws: $\bar{x} \oplus \bar{y} = \overline{x+y}$ and $\bar{x} \otimes \bar{y} = \overline{xy}$, both of which are commutative and admitting neutral element $\bar{0}$. We easily show that $(\mathbb{Z}/n\mathbb{Z}, \oplus)$ is a commutative group.

Let G and G' be two groups, and f be a homomorphism from G into G' . We assume that G is commutative. We consider in G the binary relation defined by $x\mathcal{R}y \Leftrightarrow f(x) = f(y)$. The equivalence classes are thus the classes of G modulo $\ker f$.

The canonical decomposition of f is then given as follows:

$$\begin{array}{ccc} G & \xrightarrow{f} & S \\ \downarrow s & & \uparrow i \\ G/\ker f & \xrightarrow{h} & f(G) \end{array}$$

where s is the canonical surjection from G into $G/\ker f$, the map i is the canonical injection map from $f(G)$ into G' , and h is defined by $h(\bar{x}) = f(x)$. The map h is an isomorphism of groups.

2.4.1 Cyclic groups

Definition 22 A group G is said to be cyclic (or monogenous) if there exists an element $a \in G$ such that for every $x \in G$, there exists an integer $n \in \mathbb{Z}$ with $x = a^n$. In other words, $G = \{a^n : n \in \mathbb{Z}\}$. The element a is called a generator of G , and we write $G = \langle a \rangle$.

- When the binary operation of G is denoted additively, and G is a cyclic group generated by a , then $G = \{na : n \in \mathbb{Z}\}$.

Example 23

1. The group $(\mathbb{Z}, +)$ is a cyclic. We have $\mathbb{Z} = \langle +1 \rangle = \langle -1 \rangle$.

2. The group $(\mathbb{Z}/n\mathbb{Z}, \oplus)$ is a cyclic. We have $\mathbb{Z}/n\mathbb{Z} = \langle \bar{1} \rangle$.
3. Let $n \in \mathbb{N}$ and $U_n = \{z \in \mathbb{C} : z^n = 1\}$. The group $(U_n, \times)$ is a cyclic, and $U_n = \langle \zeta_1 \rangle$, where $\zeta_1 = e^{(2\pi i/n)}$.

Let G be a group and let a be an element of G . The set $H = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of G . This subgroup is commutative even though G is not. We call it the subgroup generated by a and we write $H = \langle a \rangle$. Two cases arise:

- If for any $n \in \mathbb{Z}$ we have $a^n \neq e$, we say that a is of *infinite order*. In this case, H is infinite and the map defined by

$$\begin{array}{ccc} \mathbb{Z} & \rightarrow & H \\ n & \mapsto & a^n \end{array}$$

is a group isomorphism.

- If there exists $n \in \mathbb{Z}$ such that $a^n = e$. The least positive integer m satisfying this property is called the *order* of a , and H is necessarily finite, H reduces to

$$H = \{e, a, a^2, \dots, a^{m-1}\},$$

and we have $a^\ell = a^{\ell'}$ if and only if $\ell - \ell' \in m\mathbb{Z}$. In this case, the map defined by

$$\begin{array}{ccc} \mathbb{Z}/m\mathbb{Z} & \rightarrow & H \\ \bar{k} & \mapsto & a^k \end{array}$$

is a group isomorphism, where $\mathbb{Z}/m\mathbb{Z}$ is endowed with its canonical group structure.

We conclude that a cyclic group is either isomorphic to $\mathbb{Z}$ (when G is infinite) or to $\mathbb{Z}/m\mathbb{Z}$ for some positive integer m (when G is finite).

Here are some notes on cyclic groups:

Proposition 24

1. If a cyclic group is generated by a , then it is also generated by a^{-1} .
2. Every cyclic group is abelian.
3. Every subgroup of a cyclic group is also cyclic.
4. If an infinite cyclic group G is generated by a , then a and a^{-1} are the only generators of G .

Proof.

1. Let $G = \langle a \rangle$. Since $a = (a^{-1})^{-1}$, all the elements obtained as powers of a can also be obtained as powers of a^{-1} .
2. Let G be a cyclic group. Then $G = \langle a \rangle$ for some $a \in G$. For any $x, y \in G$ we have $x = a^n$ and $y = a^m$ for some $n, m \in \mathbb{Z}$. Therefore

$$xy = a^n a^m = a^{n+m} = a^{m+n} = a^m a^n = yx,$$

so commutativity holds.

3. Let G be a cyclic group. Then $G = \langle a \rangle$ for some $a \in G$. Let H be a subgroup of G . Every $h \in H$ can be written as $h = a^n$ for some $n \in \mathbb{Z}$. Let m be the smallest positive integer such that $a^m \in H$. Define $h_0 = a^m$. Then $H = \langle h_0 \rangle$. Indeed, $h_0 \in H$, then all the powers of h_0 are in H . Hence $\langle h_0 \rangle \subset H$. Conversely, let $t \in H$. Then $t = a^\ell$ for some $\ell \in \mathbb{Z}$. The Euclidian division of ℓ by m gives $\ell = mk + r$ with $0 \leq r < m$. We have $a^\ell \in H$ and $a^{mk} \in H$ then $a^{\ell-mk} = a^r \in H$. Since m is the smallest integer satisfying the property, necessarily $r = 0$, hence $t = h_0^k$. Thus, $t \in \langle h_0 \rangle$ and we have $H \subset \langle h_0 \rangle$.
4. Let G be an infinite cyclic group such that $G = \langle a \rangle$ for some $a \in G$. Suppose that G is generated by a^k for some $k \in \mathbb{Z}$. Then there exists $k' \in \mathbb{Z}$ such that $(a^k)^{k'} = a$. This gives $a^{kk'-1} = e$, which implies that $kk' - 1 = 0$ (since a is of infinite order). Then $kk' = 1$. Hence $k = k' = 1$ or $k = k' = -1$. Consequently, the only generators of G are a and a^{-1} .

■

2.4.2 Finite groups

Recall that a group G with a finite number of elements is called a *finite group*. This number is called the order of the group and is denoted by $|G|$.

Theorem 25 (Lagrange) *Let G be a finite group. Then the order of any subgroup of G divides the order of G .*

Proof. Let H be a subgroup of the finite group G . Consider the binary relation on G defined by:

$$\forall x, y \in G : x \mathcal{R} y \text{ if and only if } x^{-1}y \in H.$$

We can easily show that $\mathcal{R}$ is an equivalence relation. Since G is of finite order, there are a finite number of classes. On the one hand, for any $a \in G$, the class of a , denoted by aH , is given by:

$$aH = \{ah, h \in H\}.$$

On the other hand the map

$$\begin{array}{ccc} H & \longrightarrow & aH \\ t & \longmapsto & at \end{array}$$

is bijective. Therefore $|aH| = |H|$. Since the classes aH ($a \in G$) have the same cardinality as H and they form a partition of G , then the order of G is the number of classes multiplied by the order of $|H|$. This shows that the order of H divides the order of G . ■

Corollary 26 *If G is a group of finite order n , then the order of any element of G divides the order of G .*

Proof. Let ℓ be the order of a an element of G . Then ℓ is the least positive integer, such that $a^\ell = e$. Then, we can show that the set $H = \{a, a^2, a^3, \dots, a^{\ell-1}, a^\ell = e\}$ is a subgroup of G . Since this subgroup has order ℓ , thus, ℓ divides the order of G . ■

Corollary 27 *Let G be a finite group of prime order p . Then*

1. *For any $x \in G \setminus \{e\}$ we have $G = \langle x \rangle$.*
2. *The only subgroups of G are $\{e\}$ and G .*

Proof.

1. It is clear that if $x \in G \setminus \{e\}$, then the order of $\langle x \rangle$ is different from 1 and divides $|G| = p$. Since p is prime, $|\langle x \rangle| = |G|$, necessarily $\langle x \rangle = G$. In other words, G is cyclic generated by any element x belonging to G with $x \neq e$.
2. Since the order of a subgroup divides the order the group and $|G| = p$ is prime, the only subgroups of G are of order 1 and p which corresponds to $\{e\}$ and G , respectively.

■

Theorem 28 *Let $G = \langle a \rangle$ be a finite cyclic group of order n and let k be a positive integer. Then $G = \langle a^k \rangle$ if and only if $k < n$ and $\gcd(k, n) = 1$.*

Proof. Since $G = \langle a \rangle$, we have $G = \{e, a, a^2, \dots, a^{n-1}\}$. Let $k \in \mathbb{N}$ and put $H = \langle a^k \rangle$. It is clear that $H = G$ if and only if $a \in H$ (since then H must contain all powers of a and therefore all elements of G). We have $a \in H$ if and only if there exists $u \in \mathbb{Z}$ such that $a = a^{ku}$. This is equivalent to $a^{ku-1} = e$. Thus, there exists $u \in \mathbb{Z}$ such that $ku - 1 = vn$ for some $v \in \mathbb{Z}$, which is equivalent to the existence of $u, v \in \mathbb{Z}$ such that $ku + nv = 1$. This last condition is equivalent to $\gcd(k, n) = 1$. ■

2.4.3 Remarkable finite groups:

Symmetric group: Let n be a positive integer and $E = \{1, 2, \dots, n\}$. The group $S(E)$ of bijections from E into E equipped with the composition of maps, is a group of order $n!$, which we call the *symmetric group* on n elements and denote simply S_n . It is also called the group of permutations of E or $\text{Per}(E)$. Its $n!$

elements can be described as follows: each element σ of S_n can be written as a table (matrix) with 2 rows and n columns. The first row contains the elements $1, 2, \dots, n$ and the second one indicates the respective images by σ of $1, 2, \dots, n$ in the first row. For instance, the symmetric group S_3 is of order $3! = 6$. We then have $id_S = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$, $\tau_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$, $\tau_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$, $\tau_3 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$, $\sigma_1 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$, $\sigma_2 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$.

$$S_3 = \{id_E, \tau_1, \tau_2, \tau_3, \sigma_1, \sigma_2\}.$$

$\circ$	id_S	τ_1	τ_2	τ_3	σ_1	σ_2
id_S	id_S	τ_1	τ_2	τ_3	σ_1	σ_2
τ_1	τ_1	id_S	σ_1	σ_2	τ_2	τ_3
τ_2	τ_2	σ_2	id_S	σ_1	τ_3	τ_1
τ_3	τ_3	σ_1	σ_2	id_S	τ_1	τ_2
σ_1	σ_1	τ_3	τ_1	τ_2	σ_2	id_S
σ_2	σ_2	τ_2	τ_3	τ_1	id_S	σ_1

Table of multiplication of S_3

In particular, we can deduce that the group S_3 is not abelian since $\tau_1 \circ \tau_2 \neq \tau_2 \circ \tau_1$. We can also see that the group S_3 admits three subgroups of order 2 which are $H_1 = \{id_S, \tau_1\}$, $H_2 = \{id_S, \tau_2\}$ and $H_3 = \{id_S, \tau_3\}$, and one subgroup of order 3 which is $H_4 = \{id_S, \sigma_1, \sigma_2\}$. This is an example also of a non-abelian group containing abelian subgroups.

3 Rings

The ring structure involves not just one law, like the group law, but two. Of course, these two laws must be related to each other. Inspired by $\mathbb{Z}$ equipped with usual addition and multiplication, we give the definitions of a ring and derive its properties. The theory of rings appeared as a result of the research efforts of several mathematicians attempting to solve Fermat's conjecture during the 19th century.

Definition 29 Let R be a nonempty set equipped with two binary operations denoted $*$ and $\triangle$. We say that $(R, *, \triangle)$ is a ring if

1. $(R, *)$ is an abelian group,
2. The binary operation $\triangle$ is associative,
3. For any $x, y, z \in R$, $x \triangle (y * z) = (x \triangle y) * (x \triangle z)$ and $(y * z) \triangle x = (y \triangle x) * (z \triangle x)$
(We say that $\triangle$ is distributive with respect to $*$),
4. The law $\triangle$ admits a neutral element.

In addition, if Δ is commutative, we say that $(R, *, \Delta)$ is a commutative ring.

In practice, ring binary operations are denoted as $+$ and $\cdot$, respectively, and the neutral elements by 0 and 1, respectively. When there is no ambiguity, a ring $(R, +, \cdot)$ is simply denoted as R .

Example 30

1. $(\mathbb{Z}, +, \times), (\mathbb{Q}, +, \times), (\mathbb{R}, +, \times),$ and $(\mathbb{C}, +, \times)$ are commutative rings.
2. $(\mathbb{Z}/n\mathbb{Z}, \oplus, \otimes)$ is a commutative ring, where $\otimes$ is defined as $\bar{x} \otimes \bar{y} = \overline{xy}$.
3. For any nonempty set S , $(P(S), \Delta, \cap)$ is a commutative ring, where Δ denotes the symmetric difference and $\cap$ denotes the intersection of sets, respectively.
4. The set of real sequences, with addition and multiplication of sequences, is a commutative ring.
5. The set of functions from an interval I into $\mathbb{R}$, denoted by $\mathcal{F}(I, \mathbb{R})$, equipped with addition and multiplication of functions:

$$\forall x \in I, \quad (f + g)(x) = f(x) + g(x)$$

and

$$\forall x \in I, \quad (fg)(x) = f(x)g(x)$$

is a commutative ring (The neutral element for the multiplication is the constant function equal to 1).

Proposition 31 *Let R be a ring. Then we have*

1. $\forall a \in R, a0 = 0a = 0$.
2. $\forall a \in R, (-1)(a) = (a)(-1) = -a$.
3. $\forall a, b \in R, (-a)(b) = (a)(-b) = -(ab)$.
4. $\forall a, b \in R, \forall n \in \mathbb{Z}, (na)b = a(nb) = n(ab)$.

Proof.

1. We have

$$a0 = a(0 + 0) = a0 + a0.$$

Subtracting $a \cdot 0$ from both sides we get:

$$a \cdot 0 - a \cdot 0 = a \cdot 0.$$

Therefore, $a \cdot 0 = 0$. Similarly, we obtain $0 \cdot a = 0$.

2. We have

$$(-1)a + (1)a = (-1 + 1)a = 0a = 0.$$

Thus,

$$(-1)a + a = 0 \text{ gives } (-1)a = -a.$$

The second equality follows similarly.

3. We have

$$ab + (-a)b = (a + (-a))b = 0b = 0.$$

Thus,

$$ab + (-a)b = 0 \text{ gives } (-a)b = -(ab).$$

On the other hand

$$ab + a(-b) = a(b + (-b)) = a0 = 0.$$

Thus, $ab + a(-b) = 0$ which implies $a(-b) = -(ab)$.

4. We use induction on n . For $n = 1$ we have

$$(1 \cdot a)b = ab.$$

Suppose that for every n we have

$$(na)b = a(nb).$$

Then

$$((n+1)a)b = (na + a)b = (na)b + ab.$$

By induction hypothesis, Hence,

$$((n+1)a)b = a((n+1)b).$$

For negative n , we use a similar argument by induction on $-n$.

■

The n -th power of an element a in the ring R is defined by recurrence as $a^n = a^{n-1}a$ for any $n \geq 1$. It can be easily shown that for any $m, n \in \mathbb{N}$, $a^n a^m = a^{n+m}$ and $(a^n)^m = a^{nm}$.

Proposition 32 *Let R be a ring, and let $a, b \in R$ such that $ab = ba$. For any integer $n \geq 0$, we have:*

$$1. (a+b)^n = \sum_{k=0}^n C_k^n a^k b^{n-k}.$$

$$2. a^n - b^n = (a-b) \sum_{k=0}^{n-1} a^k b^{n-k-1}.$$

Proof.

1. By induction on n . For $n = 0$; we have

$$(a + b)^0 = 1 = C_0^0 a^0 b^0.$$

Suppose that the property is true for n and prove it for $n + 1$.
We have

$$\begin{aligned} (a + b)^{n+1} &= (a + b)(a + b)^n = (a + b) \sum_{k=0}^n C_n^k a^k b^{n-k} \\ &= a \sum_{k=0}^n C_n^k a^k b^{n-k} + b \sum_{k=0}^n C_n^k a^k b^{n-k} \\ &= \sum_{k=0}^n C_n^k a^{k+1} b^{n+1-k} + \sum_{k=0}^n C_n^k a^k b^{n+1-k} \\ &= a^{n+1} + b^{n+1} + \sum_{k=1}^n (C_{n+1}^k a^k b^{n+1-k}) + \sum_{k=0}^n (C_n^k a^k b^{n+1-k}). \end{aligned}$$

On the other hand we know that

$$C_{n+1}^k = C_n^{k-1} + C_n^k.$$

Then

$$(a + b)^{n+1} = \sum_{k=0}^{n+1} C_{n+1}^k a^k b^{n+1-k}.$$

2. Using a similar reasoning by induction we get the desired formula.

■

3.1 Subrings

Definition 33 Let R be a ring. A subset B of R is said to be a subring of R if:

1. B is a subgroup of the group $(R, +)$,
2. For all $x, y \in B$, $xy \in B$,
3. $1_R \in B$.

In other words, a nonempty subset B of a ring R is a subring if B is closed under both addition and multiplication, and $1_R \in B$. With these properties, $(B, +, \cdot)$ is a ring, and $1_B = 1_R$.

Example 34

1. Let R be a ring. The ring R itself is a subring of R . However, the set $\{0_R\}$, which contains only the zero element, is not a subring unless R is the zero ring.
2. The set of differentiable functions on an interval I into $\mathbb{R}$, equipped with addition and multiplication, is a subring of the set of continuous functions on I . The continuous functions on I form a subring of the set of all functions from I into $\mathbb{R}$.
3. The set of stationary real sequences is a subring of $\mathbb{R}^{\mathbb{N}}$, which is itself a subring of $\mathbb{C}^{\mathbb{N}}$.
4. The set $\mathbb{Z}[i] = \{a + ib : a, b \in \mathbb{Z}\}$ is a subring of $(\mathbb{C}, +, \times)$, and it contains $\mathbb{Z}$ as a subring.
5. The set $\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} : a, b \in \mathbb{Z}\}$ is a subring of $(\mathbb{R}, +, \times)$, where d is not a perfect square.

Proposition 35

The only subring of $(\mathbb{Z}, +, \times)$ is $\mathbb{Z}$.

Proof. Any subring of $\mathbb{Z}$ must contain 1, then for any $n \in \mathbb{N}$, it contains $n = n \cdot 1 = \underbrace{1 + 1 + \cdots + 1}_{n \text{ times}}$, and for any $n \in \mathbb{N}_-$, it also contains $n = -n(-1) = \underbrace{(-1) + (-1) + \cdots + (-1)}_{-n \text{ times}}$. Thus, it contains $\mathbb{Z}$. Then the only subring of $\mathbb{Z}$ is $\mathbb{Z}$.

■

- It is easy to show that the intersection of two subrings of a ring is a subring, but beware, the union is not in general (it has been shown that the union of two subgroups is not in general a subgroup).

3.2 Ideals

Definition 36 Let R be a commutative ring. A nonempty subset I of R is said to be an ideal of R if:

1. I is a subgroup of $(R, +)$,
2. $\forall x \in R, \forall y \in I, xy \in I$.

Example 37

1. Let R be a ring. The singleton $\{0_R\}$ and the ring R itself are ideals of R (called trivial ideals).
2. For every $n \in \mathbb{Z}$, the subset of $\mathbb{Z}$ defined by $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ is an ideal of the ring $\mathbb{Z}$ (equipped with the usual laws).

3. In the group $(\mathbb{Q}, +)$, the set $5\mathbb{Z} = \{\dots, -10, -5, 0, 5, 10, \dots\}$ is a subgroup. But $5\mathbb{Z}$ is not an ideal in the ring $(\mathbb{Q}, +, \times)$; because it is not closed under multiplication by elements in $\mathbb{Q}$. For instance $5 \in 5\mathbb{Z}$ and $\frac{1}{5} \in \mathbb{Q}$, but $1 = \frac{1}{5} \times 5 \notin 5\mathbb{Z}$ (the result remains true for $n\mathbb{Z}$ for any integer $n \geq 0$).
4. In the ring $\mathcal{F}(\mathbb{R}, \mathbb{R})$, the set of functions from $\mathbb{R}$ into $\mathbb{R}$ that vanish at 0 is an ideal.

Remark 38

1. A subring contains 1 but an ideal not necessary.
2. Let I be an ideal of a commutative ring R . If $1 \in I$, then $I = A$.

Proposition 39 Let R be a commutative ring. The intersection of two ideals of R is an ideal of R .

Proof. Let I and J be two ideals of R . We have shown that the intersection of two subgroups is a subgroup, so $I \cap J$ is a subgroup of $(R, +)$, satisfying the first condition for being an ideal. To verify the second condition, let $a \in R$ and $x \in I \cap J$. Since $x \in I$ and $x \in J$, we have $ax \in I$ (because I is an ideal), and $ax \in J$ (because J is an ideal). Therefore, $ax \in I \cap J$. This confirms that $I \cap J$ is closed under multiplication by any element from R . Hence, $I \cap J$ is an ideal of R . ■

- The union of two ideals is not generally an ideal (it has been shown that the union of two subgroups is not usually a subgroup).

3.3 Quotient ring

Let R be a commutative ring, and let I be an ideal of R . The binary relation $\mathcal{R}$ defined on R by

$$\forall x, y \in R, x \mathcal{R} y \text{ if and only if } x - y \in I$$

is an equivalence relation. We denote the quotient set by R/I .

We define on R/I two binary operations as follows:

$$\forall \bar{x}, \bar{y} \in R/I, \bar{x} \oplus \bar{y} = \overline{x + y}$$

and

$$\forall \bar{x}, \bar{y} \in R/I, \bar{x} \otimes \bar{y} = \overline{xy}.$$

Then, $(R/I, \oplus, \otimes)$ is a commutative ring called quotient ring modulo the ideal I .

Indeed, it has been proven that $(R/I, \oplus)$ is a commutative group when the group $(R, +)$ is commutative. The law $\otimes$ is commutative and associative thanks to commutativity and associativity of the second law of R . The distributivity of $\otimes$ over $\oplus$ in R/I follows from the distributivity of the second law of R over the first law of R . The neutral element is $\bar{1}$.

Example 40 For any positive integer $(\mathbb{Z}/n\mathbb{Z}, \oplus, \otimes)$ is a group and the only subring of $(\mathbb{Z}/n\mathbb{Z}, \oplus, \otimes)$ is $\mathbb{Z}/n\mathbb{Z}$ itself.

3.4 Direct product of rings

Let R and S be two rings.

- Recall that we can define on the Cartesian product

$$R \times S = \{(r, s) : r \in R, s \in S\},$$

an addition and a multiplication as follows:

$$(r_1, s_1) + (r_2, s_2) = (r_1 + r_2, s_1 + s_2) \text{ and } (r_1, s_1)(r_2, s_2) = (r_1 r_2, s_1 s_2).$$

It is straightforward to check that $R \times S$ is a ring under these binary operations.

- In the same way, we can define a ring structure on the Cartesian product $R_1 \times R_2 \times \cdots \times R_n$ of n rings $R_1, R_2, \dots, R_n$.

Example 41 $\mathbb{Z}^n$, $\mathbb{Q}^n$, $\mathbb{R}^n$, and $\mathbb{C}^n$ have canonical structures of rings induced from the structure of the rings $(\mathbb{Z}, +, \times)$, $(\mathbb{Q}, +, \times)$, $(\mathbb{R}, +, \times)$, and $(\mathbb{C}, +, \times)$.

Note that if R and S are two rings, R and S are not subrings of $R \times S$. Indeed, R can be thought as elements of the form $(r, 0)$ but it does not contain the identity element of $R \times S$.

Remarkable elements of a ring

Let R be a commutative ring.

- An element a in R is said to be *invertible* (or a unit) if there exists b in R such that $ab = 1$. The element b is denoted by a^{-1} and called the inverse of a . The set of invertible elements is denoted by R^* or $U(R)$.
- A nonzero element a in R is said to be a *divisor of zero* if there exists a nonzero element b in R such that $ab = 0$.
- An element a in R is said to be *nilpotent* if there exists $n \in \mathbb{N}$ such that $a^n = 0$. The smallest integer p which verifies this property is called the nilpotency index.
- An element a in R is said to be *idempotent* if $a^2 = a$.

Example 42

1. In any ring R , the elements 1 and -1 are invertible (In the ring $(\mathbb{Z}, +, \times)$, these are the only invertible elements).

2. In the rings $(\mathbb{Q}, +, \times)$, $(\mathbb{R}, +, \times)$, and $(\mathbb{C}, +, \times)$, all elements different from 0 are invertible.
3. The rings $(\mathbb{Z}, +, \times)$, $(\mathbb{Q}, +, \times)$, $(\mathbb{R}, +, \times)$, and $(\mathbb{C}, +, \times)$ have no zero divisors.
4. In $\mathbb{R}^{\mathbb{R}}$, the ring of maps from $\mathbb{R}$ into $\mathbb{R}$, the maps

$$f : \begin{array}{l} \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto \begin{cases} x, & \text{if } x \geq 0, \\ 0, & \text{if } x < 0 \end{cases} \end{array}$$

and

$$g : \begin{array}{l} \mathbb{R} \rightarrow \mathbb{R} \\ x \mapsto \begin{cases} 0, & \text{if } x \geq 0, \\ 1, & \text{if } x < 0 \end{cases} \end{array}$$

are zero divisors. In general, $f : \mathbb{R} \rightarrow \mathbb{R}$ is zero divisor if and only if f is not identically zero and it is equal to zero at least at one point.

5. In $\mathbb{Z} \times \mathbb{Z}$, equipped with its canonical structure as a ring, if a, b are nonzero integers, then $(a, 0)$ and $(0, b)$ are nonzero elements whose product is zero, so they are zero divisors.
6. In the ring $(\mathbb{Z}/8\mathbb{Z}, +, \times)$ The elements $\bar{0}, \bar{1}, \bar{2}, \bar{4}$ are idempotents.
7. The ring $\mathbb{Z}$ has no nontrivial idempotent elements, whereas $\mathbb{Z} \times \mathbb{Z}$ has the idempotent elements $(1, 0)$ and $(0, 1)$.

Remark 43 It is easy to show that:

1. The units of a ring R form an abelian group under multiplication $((R^*, \cdot)$ is an abelian group). For example $(\mathbb{Z}[i])^* = \{1, -1, -i, +i\}$.
2. If R and S are two rings, we have $(R \times S)^* = R^* \times S^*$.

Definition 44 Let R be a commutative ring, and let $U(R)$ be its group of units.

1. An element $b \in R$ is said to divides an element $a \in R$ if there exists $q \in R$ such that $a = bq$.
2. Two elements a and b are said to be associated when $a = bu$, with $u \in U(R)$.
3. An element is said to be irreducible or prime if it is divisible only by its associates or by the units (invertibles).

Proposition 45 1. In an integral domain, two elements a and b are associated if and only if $a \mid b$ and $b \mid a$.

2. Any associate of an irreducible element is also irreducible.

Definition 46 An integral domain is a commutative ring without zero divisors.

Example 47 The rings $(\mathbb{Q}, +, \times)$, $(\mathbb{R}, +, \times)$, and $(\mathbb{C}, +, \times)$ are integral domains.

Proposition 48 The ring $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.

Proof. Suppose n is prime and let $\bar{a}, \bar{b} \in \mathbb{Z}/n\mathbb{Z}$ with $\bar{a}\bar{b} = \bar{0}$ in $\mathbb{Z}/n\mathbb{Z}$. This means ab is divisible by n . Since n is prime, for ab divisible by n to hold, at least one of a or b must be divisible by n . Hence, there are nonzero divisors. Thus, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain. Conversely, assume $\mathbb{Z}/n\mathbb{Z}$ is an integral domain. Suppose n is not prime. Then n can be factored as $n = ab$, where $1 < a, b < n$. Consider $\bar{a}$ and $\bar{b}$ in $\mathbb{Z}/n\mathbb{Z}$. Since ab is divisible by n but neither $\bar{a}$ nor $\bar{b}$ are nonzero elements of $\mathbb{Z}/n\mathbb{Z}$ whose product is zero. Thus, $\mathbb{Z}/n\mathbb{Z}$ contains zero divisors. The presence of zero divisors contradicts the definition of an integral domain. Hence, n must be prime. ■

- If R and S are two integral domains, the ring $R \times S$ is not an integral domain. Indeed, the elements $(1_R, 0_S)$ and $(0_R, 1_S)$ are nonzero, while their product is zero.

Proposition 49 Let R be an integral domain and $a, b, c \in R$. If $ab = ac$ and $a \neq 0$ then $b = c$.

Proof. If $ab = ac$, then $ab - ac = 0$ and so $a(b - c) = 0$. Since R is an integral domain and $a \neq 0$, it follows that $b - c = 0$, hence $b = c$. ■

4 Ring homomorphisms

Definition 50 Let R and S be two rings. A map

$$f : R \rightarrow S$$

is called a ring homomorphism if:

1. For all $a, b \in R$, $f(a + b) = f(a) + f(b)$,
2. For all $a, b \in R$, $f(ab) = f(a)f(b)$,
3. $f(1_R) = 1_S$.

Example 51

1. The map

$$\begin{aligned} f : \mathbb{C} &\rightarrow \mathbb{C} \\ z &\mapsto \bar{z} \end{aligned}$$

is a ring automorphism of $(\mathbb{C}, +, \times)$.

2. The only ring homomorphism from $(\mathbb{Z}, +, \times)$ into itself is the identity map.

3. The map

$$\begin{aligned} f : \mathbb{Z} &\rightarrow \mathbb{Z}/n\mathbb{Z} \\ k &\mapsto \bar{k} \end{aligned}$$

is a ring homomorphism from $(\mathbb{Z}, +, \times)$ into $(\mathbb{Z}/n\mathbb{Z}, \oplus, \otimes)$.

4. Let p be a prime number. Then the map

$$\begin{aligned} f : \mathbb{Z}/p\mathbb{Z} &\rightarrow \mathbb{Z}/p\mathbb{Z} \\ \bar{k} &\mapsto (\bar{k})^p \end{aligned}$$

is a ring homomorphism called the Frobenius map. Indeed, we have $\bar{0}^p = \bar{0}$, $\bar{1}^p = \bar{1}$ and using binomial theorem $(a + b)^n = \sum_{k=0}^n C_k^n a^k b^{n-k}$ and the fact that p divides C_k^n when $1 \leq k \leq p-1$, the result holds.

Proposition 52 *Let $f : R \rightarrow S$ be a ring homomorphism. Then we have:*

1. $f(0_R) = 0_S$,
2. For all $a \in R$, $n \in \mathbb{Z}$, $f(na) = nf(a)$,
3. For all $a \in R$, $n \in \mathbb{N}_0$, $f(a^n) = f(a)^n$.

Proof. The assertions 1) and 2) have been proven in Proposition 18. Assertion 3) can be proved by a straightforward induction on n . ■

- If $f : R \rightarrow S$ is a ring homomorphism. It is easy to show that $\text{Im } f$ is a subring of S . However, $\ker f$ is not a subring of R since it does not contain the unit 1_R (unless S is the zero ring, i.e., $0_S = 1_S$).

Proposition 53 *Let R and S be two commutative rings, and let $f : R \rightarrow S$ be a ring homomorphism. We have the following properties hold:.*

1. For any ideal J of S , the preimage $f^{-1}(J)$ is an ideal of R .
2. $\ker f$ is an ideal of R .
3. For any ideal I of R , the direct image $f(I)$ is an ideal of the ring $f(R) = \text{Im } f$.

Note that $\text{Im } f$ is not, in general, an ideal of S .

Proof.

1. Let J be an ideal of S . We already know that $f^{-1}(J)$ is subgroup of R under addition. Let $x \in f^{-1}(J)$ and $a \in R$. We have $f(xa) = f(x)f(a)$ with $f(a) \in S$ and $f(x) \in J$. Since J is an ideal of S , $f(xa) \in J$, which implies $xa \in f^{-1}(J)$. Therefore, $f^{-1}(J)$ is an ideal of R .

2. It is enough to apply the previous result to $J = \{0_S\}$. Thus, $\ker(f) = f^{-1}(\{0_S\})$ is an ideal of R .
3. Consider I as an ideal of R . We know that the image of a subgroup under a ring homomorphism is a subgroup, then $f(I)$ is a subgroup of S under addition. Let $y \in f(I)$. Then there exists $x \in I$ such that $y = f(x)$. For any element $b \in S$ that belongs to $\text{Im } f$, there exists $a \in R$ such that $b = f(a)$. We then have $by = f(a)f(x) = f(ax)$, where $ax \in I$ (since $x \in I$ and I is an ideal). Thus, $yb \in f(I)$. This proves that $f(I)$ is an ideal of the ring $\text{Im } f$.

■

Let $f : R \rightarrow S$ be a ring homomorphism. Then we have the following canonical decomposition of f :

$$\begin{array}{ccc} R & \xrightarrow{f} & S \\ \downarrow s & & \uparrow i \\ R/\ker f & \xrightarrow{h} & f(R) \end{array}$$

In this decomposition, s is the canonical surjection which a ring homomorphism. The map i is the canonical injection which is a ring homomorphism, and h is defined by $h(\bar{x}) = f(x)$. In turn, h is an isomorphism of rings.

4.1 Characteristic of a ring

Let R be a ring and Φ the ring homomorphism defined as

$$\Phi_R : \begin{array}{l} \mathbb{Z} \rightarrow R \\ n \mapsto \begin{cases} \underbrace{1_A + 1_A + \cdots + 1_A}_{n \text{ times}} & \text{if } n > 0, \\ 0 & \text{if } n = 0, \\ \underbrace{(-1_A) + (-1_A) + \cdots + (-1_A)}_{-n \text{ times}} & \text{if } n < 0. \end{cases} \end{array}$$

$\ker \Phi_R$ is an ideal of $\mathbb{Z}$. The *characteristic of R* denoted by $\text{char}(R)$ is the defined to be the integer k such that $\ker \Phi_R = k\mathbb{Z}$.

In other words,

$$\text{char}(R) = \begin{cases} \min \{n \in \mathbb{N} : n1_A = 0\}, \\ 0 \text{ otherwise.} \end{cases}$$

Proposition 54 *Let R be a commutative ring. For any subring B of R , we have $\text{char}(B) = \text{char}(R)$.*

Proof. Let Φ_B and Φ_R be the homomorphisms defined above and i be the canonical injection of B into R . Then the following diagram

$$\begin{array}{ccc} B & \xrightarrow{i_B} & R \\ \Phi_B \searrow & & \nearrow \Phi_R \\ & \mathbb{Z} & \end{array}$$

is commutative, hence $\ker \Phi_B = \ker \Phi_R$. Therefore B and R have the same characteristic. ■

Example 55

1. $\text{char}(\mathbb{Z}) = \text{char}(\mathbb{Q}) = \text{char}(\mathbb{R}) = \text{char}(\mathbb{C}) = 0$.
2. $\text{char}(\mathbb{Z}/n\mathbb{Z}) = n$.

Proposition 56 *Let p, q , and n positive integers. Then we have*

1. $\text{char}(\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}) = pq$, if $\gcd(p, q) = 1$.
2. $\text{char}(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}) = n$.

Proof.

1. The characteristic of the direct product $\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$ is the least common multiple (lcm) of the characteristics of its components.
Thus:

$$\text{char}(\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}) = \text{lcm}(\text{char}(\mathbb{Z}/p\mathbb{Z}), \text{char}(\mathbb{Z}/q\mathbb{Z})).$$

Knowing that the characteristic of $\mathbb{Z}/p\mathbb{Z}$ is p , the characteristic of $\mathbb{Z}/q\mathbb{Z}$ is q , $\gcd(p, q) = 1$, p and q are coprime, and using the formula for lcm when p and q are coprime: $\text{lcm}(p, q) = pq$ gives,

$$\text{char}(\mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}) = \text{lcm}(p, q) = pq.$$

2. The same reasoning gives

$$\text{char}(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}) = \text{lcm}(\text{char}(\mathbb{Z}/n\mathbb{Z}), \text{char}(\mathbb{Z}/n\mathbb{Z})) = n.$$

■

4.2 The ring $(\mathbb{Z}, +, \times)$ and its arithmetic

Let $n \in \mathbb{N}$ and $x, y \in \mathbb{Z}$. We say that x is congruent to y modulo n and we write $x \equiv y \pmod{n}$ if $x - y \in n\mathbb{Z}$, which is equivalent to $x - y$ is a multiple of n (recall that we write $x \mid y$ to say x divides y).

The following properties are easy to establish.

1. If $x \equiv 0 \pmod{n}$, then x is divisible by n .
2. The relation congruent modulo n is an equivalence relation.
3. $\forall (x, y, a, b) \in \mathbb{Z}^4$, if $\begin{cases} x \equiv a \pmod{n} \\ y \equiv b \pmod{n} \end{cases}$, then $x + y \equiv a + b \pmod{n}$.

$$4. \forall (x, y, a, b) \in \mathbb{Z}^4, \text{ if } \begin{cases} x \equiv a \pmod{n} \\ y \equiv b \pmod{n} \end{cases}, \text{ then } xy \equiv ab \pmod{n}.$$

Lemma 57 (Bézout) *Let a, b be distinct positive integers. There exist $x, y \in \mathbb{Z}$ such that $xa + yb = \gcd(a, b)$.*

Proof. Let $S = \{xa + yb : x, y \in \mathbb{Z} \text{ and } xa + yb > 0\}$. It is clear that $S \subset \mathbb{N}$ and $S \neq \emptyset$ (for instance $a, b \in S$). Since $S \subset \mathbb{N}$, it has a smallest element which we denote ℓ . Let $d = \gcd(a, b)$. Since d divides both a and b , it follows that d divides any combination $xa + yb$ for $x, y \in \mathbb{Z}$. Therefore, d divides every element of S . So d divides ℓ .

Conversely, the Euclidean algorithm of a by b gives $a = q\ell + r$ with $0 \leq r < \ell$. Suppose, that $r \neq 0$ (so $r > 0$). We have $\ell = xa + yb, x, y \in \mathbb{Z}$ (since $\ell \in S$). Then $r = a - q\ell = a - q(xa + yb) = a(1 - qx) + b(-yq)$. Since $(-yq), (1 - qx) \in \mathbb{Z}$ and $r > 0$ necessarily $r \in S$. Hence $r \in S$ and $r < \ell$. This is a contradiction. Thus, $r = 0$ and ℓ divides a . In a similar way we get ℓ divides b . Then ℓ divides d . So the proof of Bézout's Lemma is complete. ■

► Note that a constructive proof of Bézout's Lemma can be derived from the Euclidean algorithm.

Theorem 58 (Wilson theorem) *An integer p strictly greater than 1 is a prime number if and only if $(p - 1)! + 1 \equiv 0 \pmod{p}$.*

Proof. First suppose that p is prime. Then each a in $\{1, 2, \dots, p - 1\}$ has an inverse $a^{-1} \in \{1, 2, \dots, p - 1\}$ modulo p ; that is $aa^{-1} \equiv 1 \pmod{p}$. This inverse is unique and it follows that $(a^{-1})^{-1} = a$. If $a = a^{-1}$ then $1 \equiv aa^{-1} = a^2 \pmod{p}$. We have seen that this necessitates $a \equiv \pm 1 \pmod{p}$ and so $a = 1$ or $a = p - 1$. In the product $(p - 1)! = 1 \times 2 \times 3 \times \dots \times (p - 2) \times (p - 1)$ we pair off each term, save for 1 and $p - 1$ with its inverse modulo p . We thus get

$$(p - 1)! \equiv 1 \times (p - 1) \equiv -1 \pmod{p}.$$

Conversely, suppose

$$(p - 1)! \equiv -1 \pmod{p}. \quad (1)$$

We have to show that p is prime. Begin by rewriting (1) as follows:

$$(p - 1)! + 1 = kp \text{ for some } k \in \mathbb{Z}.$$

Suppose

$$p = ab \quad (2)$$

with $1 \leq a \leq p$ and $1 \leq b \leq p$. If $a = p$, the factorization is trivial, so suppose $a < p$. Then a divides $(p - 1)!$ (since in this case $a \in \{1, 2, \dots, p - 1\}$) and a divides p from (2), then a divides $kp - (p - 1)! = 1$. So a divides 1. Therefore, $a = 1$. This proves that the only factorization of p is the trivial one, so p is prime. ■

- Wilson's theorem provides a characterization of prime numbers. It is therefore clearly a primality criterion. Unfortunately, its cost from an algorithmic point of view makes it unusable for testing the primality of large numbers.

Theorem 59 (Fermat's little theorem) *Let p be a prime number and a an integer not divisible by p , then*

$$a^{p-1} - 1 \equiv 0 \pmod{p}.$$

Proof. The proof of this theorem is straightforward. Let p be a prime number. We know that in the ring $\mathbb{Z}/p\mathbb{Z}$ if a is an integer not divisible by p , then the element $\bar{a} \in (\mathbb{Z}/p\mathbb{Z})^*$ that is, $\bar{a}$ is invertible. The multiplicative group $(\mathbb{Z}/p\mathbb{Z})^*$ is of order $p - 1$, we have $\bar{a}^{p-1} = \bar{1}$, which is equivalent to say $a^{p-1} - 1 \equiv 0 \pmod{p}$. ■

Fermat's little theorem has interesting consequences. A criterion of non-primality is clearly deduced.

Let φ be the map defined by

$$\begin{aligned} \varphi : \mathbb{N} &\longrightarrow \mathbb{N} \\ n &\longmapsto \text{card} \{x \in \mathbb{N} : x \leq n \text{ and } x \text{ coprime to } n\}. \end{aligned}$$

The map φ is called the *Euler's totient function*. As a map, it counts the number of positive integers less than or equal to a given integer n that are relatively prime to n .

Proposition 60 *Let p be a prime number and m, n, k positive integers. Then we have*

1. $\varphi(p^k) = p^k - p^{k-1}$.
2. If $\gcd(n, m) = 1$, then $\varphi(mn) = \varphi(m)\varphi(n)$.

Theorem 61 (Euler) *For any integer $n > 0$ and any a coprime to n , we have*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Proof. Let n be a positive integer. If $n = 1$, we necessarily have $a = 1$ and the congruence is verified. If $n \geq 2$, we can see from Bézout's theorem that the group of units of the ring $\mathbb{Z}/n\mathbb{Z}$ is

$$(\mathbb{Z}/n\mathbb{Z})^* = \{ \bar{k} : 1 \leq k \leq n \text{ and } k \text{ coprime to } n \}.$$

Hence we have

$$\text{card}(\mathbb{Z}/n\mathbb{Z})^* = \varphi(n).$$

For any integer a coprime to n , we have: $\bar{a} \in (\mathbb{Z}/n\mathbb{Z})^*$. The multiplicative group $(\mathbb{Z}/n\mathbb{Z})^*$ is of order $\varphi(n)$ then $\bar{a}^{\varphi(n)} = \bar{1}$, which is equivalent to say $a^{\varphi(n)} - 1 \equiv 0 \pmod{n}$.

■

For p prime, $\varphi(p) = p - 1$. Applying Euler's theorem for $n = p$ provides Fermat's theorem. Fermat's little theorem (or Euler's theorem) has many applications. One of them is the following criterion, known as Euler's criterion, it characterizes squares modulo p .

Theorem 62 (Euler) *For any odd prime p and any integer a , we have*

$$a^{\frac{p-1}{2}} \equiv \begin{cases} 0 \pmod{p} & \text{if } p \mid a, \\ 1 \pmod{p} & \text{if } p \nmid a \text{ and there exists } x \in \mathbb{Z} \text{ such that } x^2 \equiv a \pmod{p}, \\ -1 \pmod{p} & \text{if } p \nmid a \text{ and for any } x \in \mathbb{Z}, \text{ we have } x^2 \not\equiv a \pmod{p}. \end{cases}$$

Theorem 63 (Chinese remainder theorem) *Let p and q be relatively prime positive integers. For any integers a and b , the system of congruences*

$$\begin{cases} x \equiv a \pmod{p} \\ x \equiv b \pmod{q} \end{cases} \quad (3)$$

has a solution, and this solution is unique modulo pq .

Proof.

- Existence: Write the first congruence as an equation in $\mathbb{Z}$, say

$$x = a + py \text{ for some } y \in \mathbb{Z}.$$

Then the second congruence is the same as

$$a + py \equiv b \pmod{q}.$$

Subtracting a from both sides, we need to solve the following in y

$$py \equiv b - a \pmod{q}. \quad (4)$$

Since $\gcd(p, q) = 1$. Let p_1 be an inverse for p modulo q , so

$$pp_1 \equiv 1 \pmod{q}.$$

Multiplying through by p_1 , we have

$$y \equiv p_1(b - a) \pmod{q},$$

so

$$y \equiv p_1(b - a) + q\ell \text{ for some } \ell \in \mathbb{Z}.$$

Then

$$\begin{aligned} x &= a + py \\ &= a + p(p_1(b - a) + q\ell) \\ &= a + pp_1(b - a) + pq\ell. \end{aligned}$$

So any x satisfying the two congruences must have this form. Conversely, it is a direct verification to show that the obtained expression of x satisfies the two congruences for every $\ell \in \mathbb{Z}$. Indeed:

$$\begin{cases} pp_1(b-a) + pq\ell \equiv a + 0 + 0 \equiv a \pmod{p}, \\ a + pp_1(b-a) + pq\ell \equiv a + 1(b-a) + 0 \equiv b \pmod{q}. \end{cases}$$

- Uniqueness: Suppose $x = x_1$ and $x = x_2$ both satisfy (3), then we have

$$\begin{cases} x_1 \equiv x_2 \pmod{p} \\ x_1 \equiv x_2 \pmod{q}. \end{cases}$$

Then p divides $(x_1 - x_2)$ and q divides $(x_1 - x_2)$. Since $\gcd(p, q) = 1$, the product pq divides $x_1 - x_2$, which means

$$x_1 - x_2 \equiv 0 \pmod{pq}.$$

This shows that all solutions to the system of congruences are the same modulo pq .

■

The previous theorem can be generalized to the following:

Definition 64 Let $m_1, m_2, \dots, m_r$ be r nonzero integers. $m_1, m_2, \dots, m_r$ are said to be pairwise relatively prime if $\gcd(m_i, m_j) = 1$ for $i \neq j$, and $1 \leq i, j \leq r$.

Theorem 65 For $r \geq 2$, let $p_1, p_2, \dots, p_r$ be nonzero integers that are pairwise relatively prime. Then, for any integers $a_1, a_2, \dots, a_r$, the system of congruences

$$\begin{cases} x \equiv a_1 \pmod{p_1} \\ x \equiv a_2 \pmod{p_2} \\ \vdots \\ x \equiv a_r \pmod{p_r} \end{cases} \quad (5)$$

has a solution, and this solution is uniquely determined modulo $p_1 p_2 \cdots p_r$.

Proof. Let $N = p_1 p_2 \cdots p_r$ and for any $k \in \{1, 2, \dots, r\}$ let $y_k = \frac{N}{p_k}$, $z_k \equiv y_k^{-1} \pmod{p_k}$ (z_k exists because $p_1, p_2, \dots, p_r$ are coprime). Let

$$x = \sum_{k=1}^r a_k y_k z_k,$$

then x is a solution of the above congruence system, and modulo N , x is the only solution. Indeed, $x \equiv (a_1 y_1 z_1 + a_2 y_2 z_2 + \cdots + a_r y_r z_r) \pmod{p_k}$. This gives

$$x \equiv a_k y_k z_k \pmod{p_k} \text{ (since } y_j \equiv 0 \pmod{p_i} \text{ for } i \neq j \text{)}.$$

Then

$$x \equiv a_k \pmod{p_k} \text{ (Since } y_k z_k \equiv 1 \pmod{p_k} \text{ for any } k \in \{1, 2, \dots, r\} \text{)}.$$

Suppose now that there are two solutions x_1 and x_2 modulo N to the system of congruences (5). Then

$$p_1 \mid (x_1 - x_2), p_2 \mid (x_1 - x_2), \dots, p_r \mid (x_1 - x_2),$$

and since $p_1, p_2, \dots, p_r$ are coprime, then $p_1 p_2 \cdots p_r$ divides $x_1 - x_2$, i.e., $x_1 \equiv x_2 \pmod{p_1 p_2 \cdots p_r}$. So the solution is unique modulo $N = p_1 p_2 \cdots p_r$. ■

Example 66

1. $\begin{cases} x \equiv 4 \pmod{15} \\ x \equiv 10 \pmod{12} \end{cases}$ has as solutions $x = 34 + 180k; k \in \mathbb{Z}$.
2. $\begin{cases} x \equiv 2 \pmod{3} \\ x \equiv 3 \pmod{5} \\ x \equiv 3 \pmod{7} \end{cases}$ has as solutions $x = 23 + 105k; k \in \mathbb{Z}$.

5 Fields

A field is an algebraic structure consisting of a set equipped with two operations: addition and multiplication, which satisfy certain properties. Fields are fundamental in many areas of mathematics, particularly in solving equations and defining vector spaces...

Definition 67 *A field is any commutative ring different from zero in which any nonzero element is invertible for the second law.*

Example 68

1. The rings $(\mathbb{Q}, +, \times)$, $(\mathbb{R}, +, \times)$, and $(\mathbb{C}, +, \times)$ are fields.
2. $(\mathbb{R}, \oplus, \otimes)$ is a field, where the laws $\oplus, \otimes$ are defined as

$$\forall x, y \in \mathbb{R}, x \oplus y = x + y + 1$$

and

$$\forall x, y \in \mathbb{R}, x \otimes y = x + y + xy.$$

The neutral elements are -1 and 0 , respectively.

3. $(\mathbb{Z}/p\mathbb{Z}, \oplus, \otimes)$ is a field for any p prime. As seen before, this field is usually denoted $\mathbb{F}_p$.

A non commutative ring different from zero in which any nonzero element is invertible is called skew field or a ring with division.

Proposition 69 *Every field is an integral domain.*

Proof. Let $\mathbb{F}$ be a field and let x be a nonzero element of $\mathbb{F}$. Suppose $y \in \mathbb{F}$ such that

$$\begin{cases} xy = 0 & (1) \\ \text{or} \\ yx = 0 & (2) \end{cases}$$

Since $\mathbb{F}$ is a field and $x \neq 0$, then we compose with x^{-1} in (1) and (2) we get $y = 0$. Hence $\mathbb{F}$ is without zero divisors. ■

Proposition 70 *The only ideals of a field $\mathbb{F}$ are $\{0\}$ and $\mathbb{F}$.*

Proof. Let I be an ideal of a field $\mathbb{F}$. If $I \neq \{0\}$, then I contains at least an element $x \neq 0$, and since this element is invertible and $xx^{-1} = 1 \in I$. Thus, $I = \mathbb{F}$. ■

5.1 Subfields

Definition 71 *Let $(\mathbb{F}, +, \cdot)$ be a field. A subfield of $\mathbb{F}$ is a subring $\mathbb{K}$ of $\mathbb{F}$ such that for any nonzero element x of $\mathbb{K}$, we have $x^{-1} \in \mathbb{K}$.*

In other words, a subfield of $\mathbb{F}$ is a subring $\mathbb{K} \subset \mathbb{F}$ which is a field with respect to the induced operations.

Example 72

1. The set $\mathbb{Q}(\sqrt{2}) = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a subfield of $(\mathbb{R}, +, \times)$.
2. The set $\mathbb{Q}(i) = \{a + bi : a, b \in \mathbb{Q}\}$ is a subfield of $(\mathbb{C}, +, \times)$.

Definition 73 *Let $\mathbb{F}$ and $\mathbb{K}$ be two fields. A map $f : \mathbb{F} \rightarrow \mathbb{K}$ is called a field homomorphism if:*

1. $f(a + b) = f(a) + f(b)$ for all $a, b \in \mathbb{F}$,
2. $f(ab) = f(a)f(b)$ for all $a, b \in \mathbb{F}$,
3. $f(1_{\mathbb{F}}) = 1_{\mathbb{K}}$.

We observe that a field homomorphism is nothing more than a ring homomorphism.

Proposition 74 *A field homomorphism $f : \mathbb{F} \rightarrow \mathbb{K}$ is injective.*

Proof. The kernel of f is an ideal of $\mathbb{F}$, which must be either the zero ideal or $\mathbb{F}$ itself. The second case is not possible, otherwise f is the zero map which is not a field homomorphism. Therefore $\ker f = \{0\}$. ■

6 Exercises

Exercise 2 : We equip $\mathbb{R}$ with the commutative law given by

$$a * b = a + b - a^2 b^2.$$

1. Show that $*$ is not associative.
2. Show that
 - If $a < -\frac{1}{\sqrt[3]{\sqrt{4}}}$, then a is not invertible.
 - If $a = -\frac{1}{\sqrt[3]{\sqrt{4}}}$, then a has an inverse.
 - If $a > -\frac{1}{\sqrt[3]{\sqrt{4}}}$, then a has two inverses, except in the case $a = 0$.
3. Verify that 1 is invertible but not regular (which is possible because the law is not associative).

Exercise 3: We define on $\mathbb{R}^2$ the internal composition law $*$ as follows:

$$\forall (x, y), (x', y') \in \mathbb{R}^2, \quad (x, y) * (x', y') = \left(x + x'^{x'} + y'^{-x} \right).$$

1. Show that $(\mathbb{R}^2, *)$ is a non-abelian group.
2. Show that the subset $H = \{(x, \sinh(x)) : x \in \mathbb{R}\}$ is a subgroup of $(\mathbb{R}^2, *)$, where $\forall x \in \mathbb{R}, \sinh(x) = \frac{e^x - e^{-x}}{2}$.
3. Let the map: $f : \mathbb{R} \rightarrow \mathbb{R}^2$ defined by $f(x) = (x, \sinh(x))$.
 - (a) Show that the map f is a morphism of the groups $(\mathbb{R}, +)$ and $(\mathbb{R}^2, *)$.
 - (b) Determine $\ker f$. Deduce that f is injective.
 - (c) Is the map f surjective?

Solution:

We define on $\mathbb{R}^2$ the internal composition law $*$ as follows:

$$\forall (x, y), (x', y') \in \mathbb{R}^2, \quad (x, y) * (x', y') = \left(x + x'^{x'} + y'^{-x} \right).$$

1. Let us show that $(\mathbb{R}^2, *)$ is a non-abelian group.
For $(1, 0), (0, 1) \in \mathbb{R}^2$, we have

$$(1, 0) * (0, 1) = (1, e^{-1}) \text{ and } (0, 1) * (1, 0) = (1, e).$$

However, $e^{-1} \neq e$ which implies that $(1, 0) * (0, 1) \neq (0, 1) * (1, 0)$. Therefore the law $*$ is not commutative.

For any $(x, y), (x', y'), (z, t) \in \mathbb{R}^2$, we have

$$\begin{aligned} ((x, y) * (x', y')) * (z, t) &= (x + x'^{x'} + y'^{-x}) * (z, t) \\ &= ((x + x'^{x'} + y'^{-x})e^z + te^{-(x+x')}) \\ &= (x + x'^{x'+z} + y'^{-x+z} + te^{-x-x'}). \end{aligned}$$

On the other hand

$$\begin{aligned} (x, y) * ((x', y') * (z, t)) &= (x, y) * (x' + z, y'^z + te^{-x'}) \\ &= (x + (x'^{x'+z} + (y'^z + te^{-x'})e^{-x})) \\ &= (x + x'^{x'+z} + y'^{-x+z} + te^{-x-x'}) \\ &= ((x, y) * (x', y')) * (z, t). \end{aligned}$$

Hence the law $*$ is associative.

An element (e', e'') in $\mathbb{R}^2$ is a neutral element for $*$ if and only if for any $(x, y) \in \mathbb{R}^2$,

$$(x, y) * (e', e'') = (e', e'') * (x, y) = (x, y).$$

On the one hand we have

$$\begin{aligned} \forall (x, y) \in \mathbb{R}^2, (e', e'') * (x, y) = (x, y) &\Leftrightarrow \forall (x, y) \in \mathbb{R}^2, e' + x = x \text{ and } e''^x + ye^{-e'} = y \\ &\Leftrightarrow \forall (x, y) \in \mathbb{R}^2, e' = 0 \text{ and } e''^x + y = y \\ &\Leftrightarrow \forall (x, y) \in \mathbb{R}^2, e' = 0 \text{ and } e''^x = 0 \\ &\Leftrightarrow e' = 0 \text{ and } e'' = 0. \end{aligned}$$

On the other hand, for any $(x, y) \in \mathbb{R}^2$, we have

$$(x, y) * (0, 0) = (x, y).$$

Then $(0, 0)$ is the neutral element for the law $*$.

Let $(x, y) \in \mathbb{R}^2$, we seek an element $(x', y') \in \mathbb{R}^2$ such that

$$(x, y) * (x', y') = (x', y') * (x, y) = (0, 0).$$

We solve only one equation. Then we have

$$\begin{aligned} (x, y) * (x', y') = (0, 0) &\Leftrightarrow \begin{cases} x + x' = 0 \\ ye^{x'} + y'^{-x} = 0 \\ y'^x + ye^{-x'} = 0 \end{cases} \\ &\Leftrightarrow (x', y') = (-x, -y). \end{aligned}$$

Hence, every element $(x, y) \in \mathbb{R}^2$ admits a symmetric $(x, y)^{-1} = (-x, -y)$. We conclude from (a), (b), (c), and (d) that $(\mathbb{R}^2, *)$ is a non-abelian group.

2. The subset $H = \{(x, \sinh(x)) : x \in \mathbb{R}\}$ is a subgroup of $(\mathbb{R}^2, *)$ if and only if

$$\begin{cases} H \neq \emptyset, \\ \forall x, y \in \mathbb{R}, (x, \sinh(x)) * (y, \sinh(y))^{-1} \in H. \end{cases}$$

We have $(0, 0) = (0, \sinh(0)) \in H$, then $H \neq \emptyset$.

For all $x, y \in \mathbb{R}$, we have

$$\begin{aligned} (x, \sinh(x)) * (y, \sinh(y))^{-1} &= (x, \sinh(x)) * (-y, -\sinh(y)) \\ &= (x - y, \sinh(x) - \sinh(y)) \\ &= (x - y, e^{-x} (e^x - e^{-x}) e^{-y} - e^{-y} (e^y - e^{-y}) e^{-x}) \\ &= (x - y, e^{-x} (e^x - e^{-x}) - e^{-y} (e^y - e^{-y})) \\ &= \left(x - y, \frac{e^{-x} (e^x - e^{-x}) - e^{-y} (e^y - e^{-y})}{2} \right) \\ &= \left(x - y, \frac{e^{-x} (e^x - e^{-x}) - e^{-y} (e^y - e^{-y})}{2} \right) \\ &= (x - y, \sinh(x - y)) \in H. \end{aligned}$$

Therefore, H is a subgroup of $(\mathbb{R}^2, *)$.

3. Let $f : \mathbb{R} \rightarrow \mathbb{R}^2$ the map defined by:

$$f(x) = (x, \sinh(x)).$$

- (a) Let us show that the map f is a morphism of groups. For any $x, y \in \mathbb{R}$, we have

$$\begin{aligned} f(x) * f(y) &= (x, \sinh(x)) * (y, \sinh(y)) \\ &= (x + y, \sinh(x)e^y + \sinh(y)e^x) \\ &= \left(x + y, \frac{e^x - e^{-x}}{2} e^y + \frac{e^y - e^{-y}}{2} e^x \right) \\ &= \left(x + y, \frac{e^{x+y} - e^{-x+y}}{2} + \frac{e^{y-x} - e^{-y-x}}{2} \right) \\ &= \left(x + y, \frac{e^{x+y} - e^{-x+y} + e^{y-x} - e^{-y-x}}{2} \right) \\ &= \left(x + y, \frac{e^{x+y} - e^{-y-x}}{2} \right) \\ &= (x + y, \sinh(x + y)) \\ &= f(x + y). \end{aligned}$$

Then the map f is a morphism of groups.

(b) We have

$$\ker f = \{x \in \mathbb{R} : f(x) = (0, 0)\}.$$

For $x \in \mathbb{R}$,

$$\begin{aligned} f(x) = (0, 0) &\iff (x, \sinh(x)) = (0, 0) \\ &\iff x = 0. \end{aligned}$$

Then

$$\ker f = \{0\}.$$

Since $\ker f$ is reduced to the neutral element of $(\mathbb{R}, +)$, then the morphism f is injective.

(c) For the surjectivity of f , we remark that $\text{Im} f = H$ (H is the graph of the map $x \mapsto \sinh(x)$) and so if we take $(\alpha, \beta) \in \mathbb{R}^2 \setminus H$, then it is clear that (α, β) does not admit an antecedent by f .

For example, for $(1, 0) \in \mathbb{R}^2$ and $x \in \mathbb{R}$, we have

$$\begin{aligned} (1, 0) = f(x) &\iff (1, 0) = (x, \sinh(x)) \\ &\iff x = 1 \text{ and } \sinh(x) = 0 \\ &\iff x = 1 \text{ and } x = 0, \end{aligned}$$

which is false. Hence $(1, 0)$ does not admit an antecedent by f . Consequently, the map f is not surjective.

Exercise 4: We define on $\mathbb{R}^2$ the binary operation $\oplus$ as follows:

$$(x, y) \oplus (x', y') = (x + x', y + y' + 2xx').$$

1. Show that $(\mathbb{R}^2, \oplus)$ is an abelian group.
2. Show that $H = \{(x, x^2) : x \in \mathbb{R}\}$ is a subgroup of $(\mathbb{R}^2, \oplus)$.
3. Show that the map $\Phi : \mathbb{R} \rightarrow H$ defined by $\Phi(x) = (x, x^2)$ is an isomorphism of groups, where $\mathbb{R}$ is endowed with its canonical structure of group.

Solution:

1. The associativity and commutativity of $\oplus$ follow straightforwardly from the commutativity and associativity of the usual addition and multiplication in $\mathbb{R}$. We only need to find (e_1, e_2) such that

$$(e_1, e_2) \oplus (x, y) = (x, y) \oplus (e_1, e_2) = (x, y).$$

Since the law is commutative, we only solve one equation

$$(x + e_1, y + e_2 + 2xe_1) = (x, y).$$

Solving this, we get $e_1 = 0$ and $e_2 = 0$. Thus, the neutral element is $(0, 0)$. Let $(x, y) \in \mathbb{R}^2$. The element $(x', y') \in \mathbb{R}^2$ is the symmetric of (x, y) if and only if:

$$(x, y) \oplus (x', y') = (x', y') \oplus (x, y) = (e_1, e_2) = (0, 0).$$

Since the law $\oplus$ is commutative, we only solve one equation

$$(x, y) \oplus (x', y') = (x + x', y + y' + 2xx') = (0, 0),$$

which gives $(x', y') = (-x, -y + 2x^2)$. We conclude that $(\mathbb{R}^2, \oplus)$ is a commutative group.

2. The subset $H = \{(x, x^2) : x \in \mathbb{R}\}$ is not empty since $(0, 0) \in H$. Let $(x, x^2), (y, y^2) \in H$. We have

$$(x, x^2) \oplus (y, y^2) = (x + y, x^2 + y^2 + 2xy) = ((x + y), (x + y)^2) \in H. \text{ Hence the stability of } H \text{ under } \oplus$$

Let $(x, x^2) \in H$. The symmetric of (x, x^2) is $(-x, x^2) = (-x, (-x)^2) \in H$. We conclude that H is a subgroup of $(\mathbb{R}^2, \oplus)$.

3. Let $\Phi : \mathbb{R} \rightarrow H$ defined by $\Phi(x) = (x, x^2)$. We have:

$$\begin{aligned} \Phi(x + y) &= (x + y, (x + y)^2) \\ &= (x + y, x^2 + 2xy + y^2) \\ &= (x, x^2) \oplus (y, y^2) \\ &= \Phi(x) + \Phi(y). \end{aligned}$$

Then Φ is a homomorphism of groups. The map Φ is surjective by construction. Easily $\Phi(x) = \Phi(y)$ gives $x = y$, then Φ is injective. We conclude that Φ is an isomorphism of groups.

Exercise 5 : On $\mathbb{N}^2$, we define the operation

$$(a_1, b_1)(a_2, b_2) = (a_1 + (a_2 - b_1)^+, b_2 + (b_1 - a_2)^+)$$

where $n^+ = \sup\{n, 0\}$ for all $n \in \mathbb{N}_0$.

1. Show that this law is associative and admits a neutral element.
2. Show that $\mathbb{N}^2$ is generated by $u = (1, 0)$ and $v = (0, 1)$.
3. Show that the set $\{u, v\}$ is stable under any automorphism and deduce that the only automorphism is the identity..

Exercise 6: We define on $\mathbb{C}$ the binary operations $\oplus$ and $\otimes$ as follows: for any $z = a + ib$ and $z' = a' + ib' \in \mathbb{C}$,

$$z \oplus z' = (a + a') + i(b + b'),$$

$$z \otimes z' = aa' + i(ab' + ba').$$

1. Show that $(\mathbb{C}, \oplus, \otimes)$ is a commutative ring.
2. Determine U , the set of invertible elements in $(\mathbb{C}, \oplus, \otimes)$.
3. Determine D , the set of zero divisors in $(\mathbb{C}, \oplus, \otimes)$.
4. Let $z = a + ib \in \mathbb{C}$. For any $n \in \mathbb{N}$, let $z^{(n)} = \underbrace{z \otimes z \cdots \otimes z}_{n \text{ times}}$.
 - (a) Write $z^{(n)}$ as a function of n, a , and b .
 - (b) Solve in $\mathbb{C}$ the equation $z^{(4)} \oplus z^{(2)} = 6 + i$.

Solution:

1. The law $\oplus$ is nothing else than the usual addition in $\mathbb{C}$ and it is well-known that $(\mathbb{C}, \oplus)$ is an abelian group with 0 as neutral element. Easy calculation gives that $\otimes$ is associative, commutative, distributive over $\oplus$, and has as a neutral element 1. Therefore, $(\mathbb{C}, \oplus, \otimes)$ is a commutative ring.
2. An element $z = a + ib \in \mathbb{C}^*$ is invertible if there exists $z' = a' + ib' \in \mathbb{C}^*$, such that

$$z \otimes z' = aa' + i(ab' + ba') = 1.$$

This is equivalent to

$$\begin{cases} aa' = 1 \\ ab' + ba' = 0. \end{cases}$$

Solving this, we find $a' = \frac{1}{a}$ and $b' = -\frac{b}{a^2}$. Hence, the set of invertible elements is

$$U = \left\{ z = a + ib \in \mathbb{C}^* : a \neq 0 \right\} = \mathbb{C}^* \setminus i\mathbb{R}.$$

3. An element $z = a + ib \in \mathbb{C}^*$ is zero divisor if there exists $z' = a' + ib' \in \mathbb{C}^*$, such that

$$z \otimes z' = aa' + i(ab' + ba') = 0.$$

This is equivalent to

$$\begin{cases} aa' = 0 \\ ab' + ba' = 0. \end{cases}$$

Solving this, we get

- If $a \neq 0$, then $a' = 0$. This gives $b' = 0$ and so $z' = a' + ib' = 0$, which is impossible.
- If $a = 0$, then $b \neq 0$, and by replacing in the second equation we get $a' = 0$. We conclude that

$$D = i\mathbb{R}^*.$$

4.

(a) We prove by induction on n that if $z = a + ib \in \mathbb{C}$, we have:

$$z^{(n)} = a^n + i(na^{n-1}b) \quad (n \geq 1).$$

(b) Using the previous question, the equation $z^{(4)} \oplus z^{(2)} = 6 + i$ is equivalent to

$$\begin{cases} a^4 + a^2 = 6 \\ 4a^3b + 2ab = 1. \end{cases}$$

Solving this, we obtain as solutions

$$z_1 = +\sqrt{2} + i\frac{1}{10\sqrt{2}} \text{ and } z_2 = -\sqrt{2} - i\frac{1}{10\sqrt{2}}.$$

Exercise 7: Let R be a ring in which every element x satisfies $x^2 = x$.

1. Give examples of such rings.
2. Show that for any $a \in R$, $2a = 0$.
3. Deduce that R is commutative.
4. Show that R cannot have exactly three elements.
5. Assume that R is finite and has cardinality greater than 2. Show that R has zero divisors.
6. Show that if $|R| = 4$, then R is unique up to isomorphism.
7. Show that if R is finite, then its cardinality is a power of 2.

Solution:

Let R be a ring with unity in which every element x satisfies $x^2 = x$.

1. The rings $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ with their canonical structures satisfy the condition $\forall x \in R, x^2 = x$.
2. On the one hand:

$$(a + a)^2 = a^2 + 2a + a^2 = a + 2a + a = 4a \tag{6}$$

On the other hand:

$$(a + a)^2 = (2a)^2 = 2a \tag{7}$$

Equating 6 and 7 gives $4a = 2a$, hence $2a = 0$.

3. On the one hand:

$$(a + b)^2 = a^2 + ab + ba + b^2 = a + ab + ba + b.$$

On the other hand:

$$(a + b)^2 = a + b.$$

Therefore:

$$a + ab + ba + b = a + b.$$

which simplifies to:

$$ab + ba = 0$$

Thus:

$$ab = -ba.$$

From part 2, we have $+1 = -1$. Therefore:

$$ab = ba.$$

4. Suppose R is a ring with cardinality 3. Thus, $R = \{0, 1, a\}$ with $a \neq 0$ and $a \neq 1$. Then $a + 1 = 0$ gives $a = -1 = 1$, which is impossible. Similarly, $a + 0 = 0$ implies $a = 0$, which is also impossible. Finally, $a + 1 = a$ implies $1 = 0$, which is again impossible. We conclude that R cannot have cardinality 3.
5. Let $R = \{0, 1, a, b, \dots\}$, where $a \neq b$ and a, b are different from 0 and 1. Since R is commutative and for any $x \in R$, $x^2 = x$, we have:

$$(a + b)(ab) = a^2b + bab = a^2b + b^2a = 2ab = 0.$$

- If $a + b \neq 0$ and $ab \neq 0$, the zero divisors are $a + b$ and ab .
- If $ab = 0$, the zero divisors are a and b since their product is zero and both are nonzero.
- The case $a + b = 0$ is not possible since it implies $a = b$. We conclude that R has zero divisors.

6. If $|R| = 4$, then the possible structures of R up to isomorphism is known and is indeed unique. It is $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

If R is finite, then its cardinality must be a power of 2. This follows from the fact that in such rings, every element is idempotent, and the number of elements in R must be a power of 2.

Exercise 8 :

Let $(R, +, \cdot)$ be a commutative ring and I, J ideals of R . The *radical* of I is defined by $\sqrt{I} := \{x \in R : \exists n \in \mathbb{N} : x^n \in I\}$.

1. Show that $\sqrt{I}$ is an ideal of R and $I \subset \sqrt{I}$.
2. Show that if $I \subset J$, then $\sqrt{I} \subset \sqrt{J}$.
3. Deduce that $\sqrt{\sqrt{I}} = \sqrt{I}$.
4. Show that $\sqrt{I} \cap \sqrt{J} = \sqrt{I \cap J}$ and $\sqrt{I + J} \supseteq \sqrt{I} + \sqrt{J}$.
5. Application: Let $R = \mathbb{Z}$ and $I = 3648\mathbb{Z}$. Find $\sqrt{I}$.

Solution

Let x and y be two elements of $\sqrt{I}$: $x^n \in I$, $y^m \in I$. There exists p such that $(x - y)^p \in I$; we can choose $p = m + n$:

$$\begin{aligned} (x - y)^{m+n} &= \sum_{p=0}^{m+n} (-1)^{m+n-p} \binom{m+n}{p} x^p y^{m+n-p} = \sum_{p=0}^n + \sum_{p=n+1}^{m+n} \\ &= y^m \left[\sum_{p=0}^n (-1)^{m+n-p} \binom{m+n}{p} x^p y^{n-p} \right] \\ &\quad + x^n \left[\sum_{p=n+1}^{m+n} (-1)^{m+n-p} \binom{m+n}{p} x^{p-n} y^{m+n-p} \right] \end{aligned}$$

The expressions inside the brackets are elements of A . Since $x^n \in I$ and $y^m \in I$, both terms on the right-hand side are elements of I , and their sum is also in I . Thus, the set $\sqrt{I}$ is an additive subgroup. It is also closed under multiplication: if $a \in A$ and $x \in \sqrt{I}$, then $(ax)^n = a^n x^n \in I$, so $ax \in \sqrt{I}$. Therefore, we have shown that $\sqrt{I}$ is an ideal. Moreover, every element of I belongs to $\sqrt{I}$.

Let $x \in \sqrt{I}$; there exists n such that $x^n \in I$, so $x^n \in J$, which implies $x \in \sqrt{J}$. Thus, $\sqrt{I} \subset \sqrt{J}$. From the previous inclusion $I \subset \sqrt{I}$, we also have $\sqrt{I} \subset \sqrt{\sqrt{I}}$.

Conversely, let $y \in \sqrt{\sqrt{I}}$; there exists p such that $y^p \in \sqrt{I}$, and there exists q such that $(y^p)^q \in I$, so $y \in \sqrt{I}$. This implies the opposite inclusion $\sqrt{\sqrt{I}} \subset \sqrt{I}$, hence the equality $\sqrt{\sqrt{I}} = \sqrt{I}$.

Exercise 9 : For any nonnegative integer n , define $F_n = 2^{2^n} + 1$.

1. Show that for any nonnegative integer n and any positive integer k , F_n divides $F_{n+k} - 2$.
2. Let n , m , and d be three nonnegative integers such that $n \neq m$. Show that if d divides both F_n and F_m , then $d = 1$.

Solution:

1. We know that for all $m \in \mathbb{N}$ and for all $(a, b) \in \mathbb{Z}^2$, $b^m - a^m = (b - a)(b^{m-1} + b^{m-2}a + \dots + ba^{m-2} + a^{m-1})$, then

$$b - a \text{ divides } b^m - a^m.$$

Let $n \in \mathbb{N}_0$ and $k \in \mathbb{N}$. Applying this result with $a = 1$, $b = 2^{2^n}$, and $m = 2^k$, we obtain

$$2^{2^n} + 1 \text{ divides } \left((2^{2^n})^{2^k} - 1 \right),$$

that is,

$$F_n \text{ divides } F_{n+k} - 2$$

(with $a^m = 1$ since $k \neq 0$).

2. Following 1) if d divides F_n and d divides F_{n+k} , then d divides 2 but for any n , F_n is odd. Then necessarily $d = 1$.

Exercise 10: Consider the field $\mathbb{Z}/p\mathbb{Z}$ where p is prime.

1. Show that the map $\varphi : x \mapsto x^p$, from $\mathbb{Z}/p\mathbb{Z}$ to itself, is an automorphism of fields.
2. Show that the only automorphism of $\mathbb{Z}/p\mathbb{Z}$ is the identity map.
3. Deduce Fermat's little theorem: if a is an integer and p is a prime number, then:

$$a^p \equiv a \pmod{p}.$$

Solution:

1. Since the characteristic is p , we have:

$$(x + y)^p = x^p + y^p,$$

so the map φ is compatible with addition. Since $\mathbb{Z}/p\mathbb{Z}$ field so the commutativity, it is also compatible with multiplication. Hence, φ is an endomorphism of the field. Since the image of φ is not reduced to $\{0\}$ (because $\varphi(1) = 1$), φ is then an automorphism.

2. Let φ_1 be an automorphism of the field. Any element x can be written as $x = k \cdot 1$, where $k \in \{0, 1, \dots, p-1\}$. Thus, we have:

$$\varphi_1(x) = \varphi_1(1 + 1 + \dots + 1) = k\varphi_1(1) = k \cdot 1 = x.$$

Hence, φ_1 is the identity automorphism. Thus, we conclude that the only automorphism of $\mathbb{Z}/p\mathbb{Z}$ is the identity map.

3. Therefore, for all $x \in \mathbb{Z}/p\mathbb{Z}$, we have:

$$x^p = x.$$

Considering the elements of the field as equivalence classes modulo p , we deduce Fermat's little theorem.

Exercise 11:

1. Show that any group of prime order is cyclic.
2. Let $n \geq 2$ be an integer. Let $1 \leq k \leq n$ and $\varphi_k : \frac{\mathbb{Z}}{n\mathbb{Z}} \longrightarrow \frac{\mathbb{Z}}{n\mathbb{Z}}$
 $\bar{x} \longmapsto \overline{kx}$.
 - (a) Show that φ_k is a group endomorphism.
 - (b) Let $\varphi : \mathbb{Z}/n\mathbb{Z} \longrightarrow \mathbb{Z}/n\mathbb{Z}$ be a map. Show that φ is a group endomorphism if and only if there exists an integer $k \in \{1, \dots, n\}$ such that $\varphi = \varphi_k$.
 - (c) Show that φ_k is a ring endomorphism if and only if n divides $k(k-1)$.
 - (d) What are the ring endomorphisms of $\mathbb{Z}/4\mathbb{Z}$?
 - (e) What are the ring endomorphisms of $\mathbb{Z}/6\mathbb{Z}$?
3. Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. We define on G the addition $\oplus$ as :

$$\forall (\bar{a}, \bar{b}), (\bar{c}, \bar{d}) \in G : (\bar{a}, \bar{b}) \oplus (\bar{c}, \bar{d}) = (\overline{a+c}, \overline{b+d}).$$

- (a) What is the cardinality of G ? List the elements of G , and draw the addition table of G .
 - (b) Can we say that all the proper subgroups of G are cyclic?
 - (c) Give all the proper subgroups of $(G, \oplus)$.
 - (d) Justify why $(G, \oplus)$ is not isomorphic to $(\mathbb{Z}/4\mathbb{Z}, +)$.
 - (e) Let H be a group such that all its proper subgroups are cyclic. Can we say that H is cyclic ?
4. Now we define on G the multiplication $\odot$ as:

$$\forall (\bar{a}, \bar{b}), (\bar{c}, \bar{d}) \in G : (\bar{a}, \bar{b}) \odot (\bar{c}, \bar{d}) = (\overline{ac}, \overline{bd}).$$

- (a) Draw the multiplication table of G .
 - (b) Knowing that $(G, \oplus, \odot)$ is a commutative ring with $(\bar{1}, \bar{1})$ as unity, determine :
 - D the set of the zero divisors.
 - U the set of invertible elements.
 - N the set of nilpotent elements.
 - I the set of idempotent elements.
 - (c) What is the characteristic of G ?
 - (d) is $(G, \oplus, \odot)$ a field ?
5. Let $n, m \geq 2$ be two integers. In this part, $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ is equipped with the addition $\boxplus$ defined by:

$$\forall (\bar{a}, \widehat{b}), (\bar{c}, \widehat{d}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z} : (\bar{a}, \widehat{b}) \boxplus (\bar{c}, \widehat{d}) = (\overline{a+c}, \widehat{b+d}).$$

- (a) Show that the order of $(\bar{1}, \hat{1})$ in $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is $\text{lcm}(n, m)$.
(b) Deduce that $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is isomorphic to $(\mathbb{Z}/nm\mathbb{Z}, +)$ if and only if $\gcd(n, m) = 1$.

Solution:

1. Let p be a prime and G a group such that $|G| = p$. Then G contains more than one element. Let $g \in G$ such that $g \neq e_G$. Then $\langle g \rangle$ contains more than one element. Since $\langle g \rangle$ is a subgroup of G , by Lagrange's theorem, $|\langle g \rangle|$ divides p . Since $|\langle g \rangle| > 1$, $|\langle g \rangle| = p = |G|$. Hence, $\langle g \rangle = G$. It follows that G is cyclic.

2.

- (a) Let $\bar{x}, \bar{y} \in \mathbb{Z}/n\mathbb{Z}$. We have

$$\begin{aligned}\varphi_k(\bar{x} + \bar{y}) &= \varphi_k(\overline{x + y}) \\ \overline{k(x + y)} &= \overline{kx + ky} = \overline{kx} + \overline{ky} = \varphi_k(\bar{x}) + \varphi_k(\bar{y}).\end{aligned}$$

Then φ_k is a group endomorphism.

- (b) Suppose that $\varphi = \varphi_k$ for some $1 \leq k \leq n$. By the previous question φ is a group endomorphism. Conversely, if φ is a group endomorphism then for any $\bar{x} \in \mathbb{Z}/n\mathbb{Z}$,

$$\varphi(\bar{x}) = \varphi(\underbrace{\bar{1} + \bar{1} + \cdots + \bar{1}}_{x \text{ times}}) = x\varphi(\bar{1}) = \overline{kx},$$

where $\bar{k} = \varphi(\bar{1})$, with $1 \leq k \leq n$.

- (c) φ_k is a ring endomorphism if and only if

$$\forall \bar{x}, \bar{y} \in \mathbb{Z}/n\mathbb{Z}, \varphi_k(\overline{xy}) = \varphi_k(\bar{x})\varphi_k(\bar{y})$$

$\iff$

$$\forall \bar{x}, \bar{y} \in \mathbb{Z}/n\mathbb{Z}, \overline{kxy} = \overline{k^2xy}$$

$\iff$

$$k^2 = k \pmod{n}$$

$\iff$

$$n \text{ divides } k(k - 1).$$

- (d) Using the previous result, the endomorphisms of $\mathbb{Z}/4\mathbb{Z}$ are φ_1, φ_4 , and the endomorphisms of $\mathbb{Z}/6\mathbb{Z}$ are $\varphi_1, \varphi_3, \varphi_4, \varphi_6$.

3.

(a) We have

$$|G| = |\mathbb{Z}/2\mathbb{Z}| \times |\mathbb{Z}/2\mathbb{Z}| = 2 \times 2 = 4$$

and

$$G = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1}), (\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}.$$

The addition table of G is the following

$\oplus$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$
$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$

(b) By Lagrange's theorem, the order of a subgroup must divide the order of the group, then the proper subgroups of G are of order 2. The subgroups of order 2 are cyclic (thanks to question 1). Then we conclude that all the proper subgroups of G are cyclic.

(c) The proper subgroups of G are $F_1 = \langle (\bar{0}, \bar{1}) \rangle = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1})\}$, $F_2 = \langle (\bar{1}, \bar{0}) \rangle = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{0})\}$ and $F_3 = \langle (\bar{1}, \bar{1}) \rangle = \{(\bar{0}, \bar{0}), (\bar{1}, \bar{1})\}$.

(d) Suppose ψ is a group homomorphism from $(\mathbb{Z}/4\mathbb{Z}, +)$ to $(G, \oplus)$, then $\psi(\bar{0}) = (\bar{0}, \bar{0})$ and $\psi(\bar{2}) = \psi(\bar{1}) + \psi(\bar{1}) = 2\psi(\bar{1}) = (\bar{0}, \bar{0}) = \psi(\bar{0})$ (since any element different from $(\bar{0}, \bar{0})$ is of order 2 in G) therefore ψ cannot be injective. We conclude then there is no group isomorphism between $(\mathbb{Z}/4\mathbb{Z}, +)$ and $(G, \oplus)$.

(e) When H is a group such that all its proper subgroups are cyclic we cannot say that H is cyclic. The group G is an example of non cyclic group whose proper subgroups are cyclic.

4.

(a) The multiplication table of G is the following:

$\otimes$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$
$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$
$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$
$(\bar{1}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{0})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{0})$
$(\bar{1}, \bar{1})$	$(\bar{0}, \bar{0})$	$(\bar{0}, \bar{1})$	$(\bar{1}, \bar{0})$	$(\bar{1}, \bar{1})$

(b) We have $D = \{(\bar{1}, \bar{0}), (\bar{0}, \bar{1})\}$, $U = \{(\bar{1}, \bar{1})\}$, $N = \{(\bar{0}, \bar{0})\}$ and $I = \{(\bar{0}, \bar{0}), (\bar{0}, \bar{1})(\bar{1}, \bar{0}), (\bar{1}, \bar{1})\}$.

(c) The characteristic of G is 2, since G is not the zero ring and $(\bar{1}, \bar{0}) + (\bar{1}, \bar{0}) = (\bar{0}, \bar{0})$, $(\bar{0}, \bar{1}) + (\bar{0}, \bar{1}) = (\bar{0}, \bar{0})$, $(\bar{1}, \bar{1}) + (\bar{1}, \bar{1}) = (\bar{0}, \bar{0})$.

(d) $(G, \oplus, \odot)$ is not a field since $(\bar{0}, \bar{1})$ is not invertible.

5.

- (a) The order of $(\bar{1}, \hat{1})$ in $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is the least $k \in \mathbb{N}^*$ such that $k(\bar{1}, \hat{1}) = (\bar{0}, \hat{0})$. If $k \in \mathbb{N}^*$ with $k(\bar{1}, \hat{1}) = (\bar{0}, \hat{0})$ then $(\bar{k}, \hat{k}) = (\bar{0}, \hat{0})$ then n divides k and m divides k , hence k is a common multiple of n and m . Since $\text{lcm}(n, m)$ is the least common multiple of n and m , then $\text{lcm}(n, m)$ is the least $k \in \mathbb{N}^*$ satisfying $k(\bar{1}, \hat{1}) = (\bar{0}, \hat{0})$. Then the order of $(\bar{1}, \hat{1})$ is $\text{lcm}(n, m)$.
- (b) The order of the group $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is nm . Suppose

$$\gcd(n, m) = 1.$$

Then $\text{lcm}(n, m) = nm$. Thanks to the previous question, the element $(\bar{1}, \hat{1})$ is of order nm which is the order of the group $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$. Then the group $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is cyclic and $(\bar{1}, \hat{1})$ is a generator. Knowing that any two cyclic groups of the same order are isomorphic then $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is isomorphic to $(\mathbb{Z}/nm\mathbb{Z}, +)$. Conversely, let $l := \text{lcm}(n, m)$. Suppose that

$$d = \gcd(n, m) \geq 2,$$

and then there exist two natural numbers m' and n' such that $m = dm'$ and $n = dn'$ with $\gcd(n', m') = 1$. We know that $\text{lcm}(n, m) \times \gcd(n, m) = mn$. Hence, $l = dm'n' = m'n = mn' < mn$. Let $(\bar{x}, \hat{y}) \in \mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$, we have

$$l(\bar{x}, \hat{y}) = \underbrace{(\bar{x}, \hat{y}) \boxplus (\bar{x}, \hat{y}) \boxplus \cdots \boxplus (\bar{x}, \hat{y})}_{l \text{ times}} = (l\bar{x}, l\hat{y}) = (\bar{0}, \hat{0}).$$

Thus, any element of $\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}$ has an order strictly less than mn . This contradicts the fact that $(\mathbb{Z}/n\mathbb{Z} \times \mathbb{Z}/m\mathbb{Z}, \boxplus)$ is cyclic of order mn . Therefore $\gcd(n, m) = 1$

Exercise 12: Let S be a nonempty set, and let $\mathcal{P}(S)$ denote the power set of S . Recall that for a subset A of B , $\bar{A}$ denotes the complement of A in B . Define an internal composition law $*$ as follows:

$$\forall A, B \in \mathcal{P}(S) : A * B = \bar{A} \cup \bar{B}.$$

1. Is the law $*$ commutative? is it associative? Does it have a neutral element?
2. Express $A \cap B$, $A \cup B$, S , $\emptyset$ solely in terms of A , B , and $*$.
3. Simplify the expression $[(A * A) * (A * A)] * [(A * A) * (A * A)]$.
4. Solve the following equations in $\mathcal{P}(S)$: $X * S = S$, $X * \emptyset = \bar{X}$, and $X * S = \emptyset$.